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Developing Standards for Functionality and Usability of the DELI Calculator

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Abstract

DELI (“fooD rEcipe environmental Impact”) Calculator is a comprehensive food recipe calculator designed to evaluate multiple environmental impact categories, developed by Paula Alejandra Barco Alzate. Building on this foundational work, this thesis project conducts preliminary research to create an effective and user-friendly interface for the DELI Calculator, focusing on improving its flexibility, accessibility, and ability to contextualize data results while incorporating user feedback. A systematic review of available online food impact calculators was conducted, followed by a user feedback survey to evaluate their utility and usability and collect user preferences. These insights were used to define standards for the functionality and design features of the DELI Calculator. With these standards established, the Master’s thesis next semester will focus on developing a tool that empowers users to make informed and sustainable food choices.

Abbreviations

AI	Artificial Intellegence
UI	User Interface
LCA	Life Cycle Assessment
GHG	Greenhouse gas

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1 Introduction

1.1 Motivation

The global food production system is exerting significant pressure on the Earth's systems, contributing to the transgression of planetary boundaries [1]. The concept of “planetary boundary” was first introduced in 2009 [2] to define a safe operating space and limits for humanity, aimed at preserving the stability and resilience of the Earth's system. There are nine boundaries in total, including climate change, biosphere integrity, and land-system change. Recent research from 2023 [3] found that six of these nine boundaries have already been crossed, signaling that the planet is now well beyond the safe zone for human activity. Agriculture, as a dominant force behind many environmental threats such as climate change, biodiversity loss, and degradation of land and freshwater [4], is a major driver pushing the environment beyond these planetary boundaries [5]. This highlights the unsustainability of current food consumption patterns, both from a health and environmental perspective [6].

Environmental impact categories, such as Greenhouse gas (GHG) emissions, land use, and eutrophication among others, quantify the environmental burden of food production. GHG emissions, expressed as CO₂-equivalents (CO₂-eq), compare emissions from various GHGs, such as carbon dioxide, methane, and nitrous oxide, based on their global warming potential (GWP) [7]. Land use measures the area required for food production [8], while eutrophication quantifies nutrient pollution in water bodies, causing harmful ecological effects [9]. These impact categories are essential for assessing and comparing the environmental impacts of different foods and diets.

Current patterns of food consumption are unsustainable and significantly contribute to the environmental challenges [6]. Tools that provide actionable knowledge and support are then needed to address this issue and empower consumers to make informed and sustainable food choices. This project focuses on establishing standards for the functionality and usability of a final environmental impact calculator for food recipes, the DELI Calculator.

1.2 Project background

1.2.1 RAINFOREST project

The RAINFOREST project is an EU project funded by the EU's Horizon Europe which aims to reduce biodiversity impacts in European food and biomass chains [10]. It focuses on closing knowledge gaps and developing sustainable practices through stakeholder collaboration. Key objectives include enhancing biodiversity impact models, co-developing just and viable transformative change pathways, and communicating results and tools through clear visualizations to a diverse audience [11]. RAINFOREST's tools and findings are designed to support diverse audiences and drive practical policy changes. Among its key outputs is the development of an environmental impact calculator for food recipes, aimed at helping individuals and policymakers make more sustainable food choices. By focusing on reducing biodiversity impacts in food systems, RAINFOREST laid the groundwork for exploring tools to assess these

effects on a more granular level.

1.2.2 Barco Alzate’s preliminary work

As part of the RAINFOREST project, Paula Alejandra Barco Alzate authored the specialization project *Life Cycle Databases for Food Sustainability Analysis* in 2023 and later completed her master’s thesis in spring 2024, titled *A comprehensive recipe calculator for evaluating multiple environmental impact categories – A first prototype* [12].

The specialization project work first reviewed available open-access Life Cycle Assessment (LCA) databases for food, selecting Poore and Nemecek (2018) [13] as the data source for the future tool to calculate the environmental impacts of food recipes. Building on this, her thesis later developed a comprehensive food recipe calculator prototype, named DELI Calculator (“fooD rEcipe environmental Impact”), with the main research question: What are the environmental impacts of selected recipes from a broader impact category perspective?. More detailed information about the thesis paper is publicly available on NTNU Open [12].

Barco Alzate’s thesis contributed preliminary work focusing on automating the retrieval of LCA scoring, without developing the UI. This means that to use this tool to calculate the environmental impacts of the food, users must manually install the Python package from Barco Alzate’s GitHub repository, follow the tutorial, and run the script using a Python interpreter to view the result. The tool provides four main visualizations of the results: a Spider web chart comparing the environmental performance of different recipes, a 100% stacked bar chart showing impact distribution per recipe broken down into ingredients, a bar chart, and a stacked bar chart per impact category. Figure 1.1 shows the Spider web chart which compares recipes with different protein options.

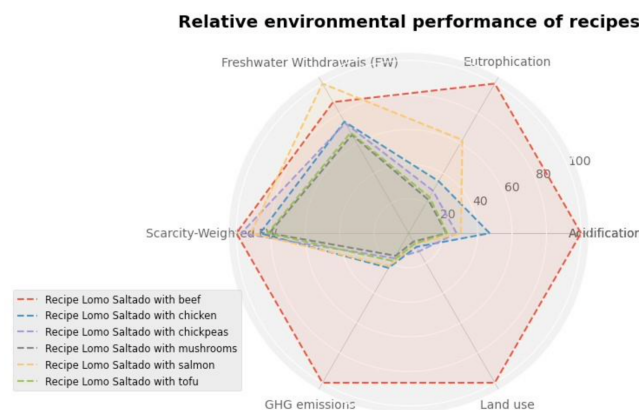


Figure 1.1: Result visualization of the DELI Calculator of 6 different Lomo Saltado recipes, screenshot from Barco Alzate’s thesis [12]

1.3 Key concepts

Building on Barco Alzate’s work, there is now a need to develop a UI to enhance the tool’s flexibility and accessibility. This subchapter introduces two key concepts, UI and Wireframing.

1.3.1 User Interface (UI)

UI or just the *interface* is part of the computer system with which users interact to undertake their tasks and achieve their goals [14]. This project focuses on the data visualization of the calculator's results along with the tool's utility and usability. The tool's utility describes the tool's usefulness which involves the interface functionality including, for example, the ingredients selection process. Usability is defined as "the extent to which specified users can use a system, product or service to achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use" according to the ISO 9241 standard [15]. It describes "how easy it is to use" the tool involving the tool's interface design including the layout design, interface aesthetics, learnability of the results, and subjective satisfaction [16]. As a good UI design can encourage an easy, natural, and engaging interaction between a user and a system, it can make the users carry out their required tasks with convenience [14].

1.3.2 Mock up wireframing

After gathering and defining the tool's usability and utility requirements, the UI of the tool can be designed [14]. A *wireframe* is a rough visual outline of a proposed application [16], serving as a mock up during the UI design process. It provides a basic structural framework of the application page, consisting of simple lines and boxes that indicate the placement of application element [17] and a more realistic prototype is developed from a wireframe [18]. Figure 1.2 below provides an example of a wireframe [16].

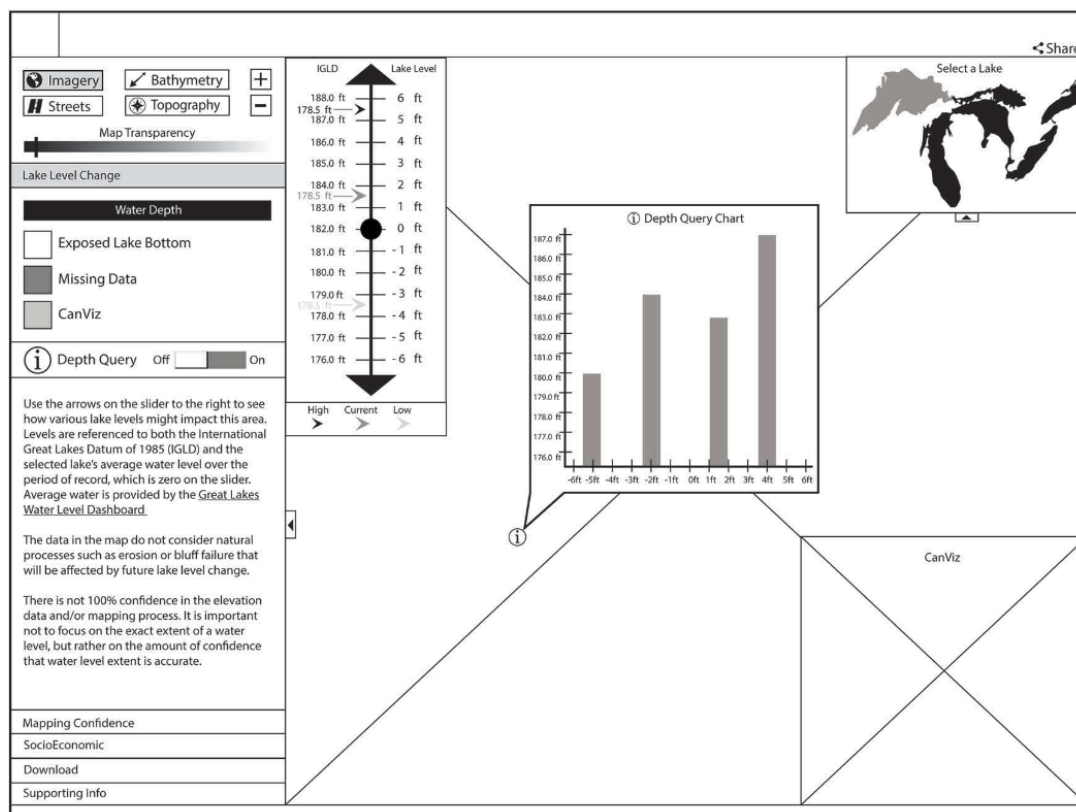


Figure 1.2: Example of a wireframe, screenshot from the article [16]

According to the ISO/IEC 14759 standard, a mock up is defined as “a provisional product that cannot be piloted by users, is not intended to evolve into a fully operational product and may be thrown away once its objective has been achieved” [19]. This differs from a *prototype*, also defined by the ISO/IEC 14759 standard, as “something that can be piloted, is developed as a part of the target product and may evolve into an operational product”. Essentially, a prototype is a functional, testable product, while a mock up provides a non-functional visual representation of the target deliverable. The categorization of mock up and prototype models can be further refined based on four possible attributes: illustrative, functional, demonstrative, and operational. Table 1.1 presents these terms and definitions, as well as distinctions between mock ups and prototypes.

Table 1.1: Terms and definitions for the categorization of mock-up and prototype models, obtained from the technical report [19]

Term	Definition
Mock up	A provisional product that cannot be piloted by users.
Prototype	A usable product that can be piloted
Illustrative	A non-functional product
Functional	A product capable of performing computations
Demonstrative	A product which proves the relevance of a solution
Operational	A product that functions in real conditions of operations

Combining these three attribute pairs of terms results in the eight possible types of products shown in Figure 1.3. This project focuses on defining standards to design a wireframe, a visual layout containing only design and layout elements, without any clickable or interactive functionality, categorized as a demonstrative, illustrative mock up.

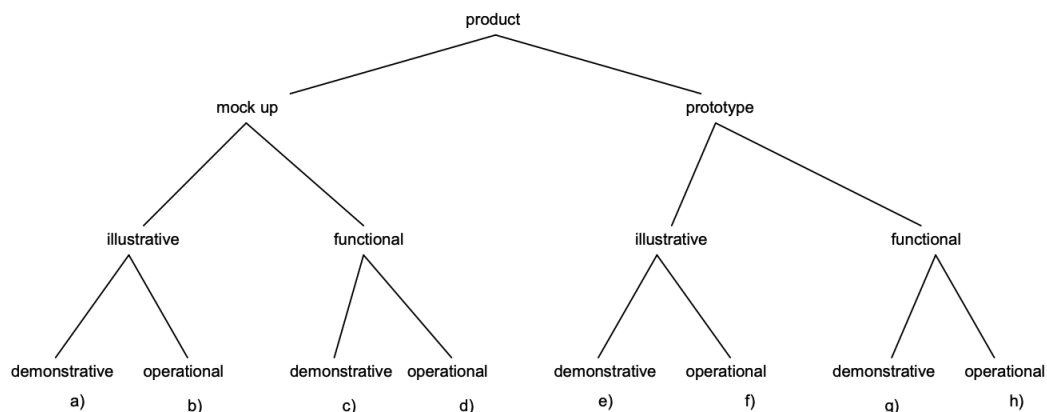


Figure 1.3: Three pairs of terms, leading to eight possible types of products, screenshot from the technical report [19]

1.4 Goal and scope

The primary goal of this project is to conduct preliminary research to develop an effective UI for the DELI calculator, emphasizing evaluating data contextualization and gathering user feedback. The research question asked to achieve this goal is “How can the impact results from the calculator be presented effectively to improve user comprehension?”.

To support this goal, the following secondary objectives were identified.

- Review existing impact calculators for food recipes that are available.
- Conduct a user feedback survey to gather insights on preferences, desired features, and usability aspects for the future design.
- Define standards for the functionality and usability of the DELI calculator’s interface, including key elements and features.

The final results of this project will establish a foundation for designing the DELI calculator’s UI in the next phase of development. By screening existing tools and analyzing user feedback, this project will define essential design elements and user interface requirements, setting the stage for the detailed wireframing and implementation planned for the Master’s thesis next semester.

2 Methodology

First, available online environmental impact calculators were identified and evaluated through a systematic review. By conducting a user feedback survey, available tools were analyzed based on their utility and usability to further define standards on functionality and design features of the final DELI Calculator.

2.1 Systematic review

Existing environmental impact calculator tools for food recipes were reviewed to assess state of the art and to gather inspiration for designing the UI of the final product.

2.1.1 Definition of an available tool

First, the following set of criteria was defined to classify a tool as “available”. In this context, a tool refers to an environmental impact calculator for food recipes.

1. The tool must be web-based, aligning with the format of the final prototype, which will also be a website. This ensures consistency and can facilitate comparative evaluation.
2. The tool must be free of charge and accessible without booking a demonstration.
3. The tool must allow users to calculate the environmental impact of a food dish, offering the ability to customize and modify the ingredients used.

2.1.2 Google Search engine

The Google search engine was used to identify publicly accessible environmental impact calculators for food recipes online. For the search, the following two sets of keywords were used during the period from September 2nd to September 22nd, 2024.

1. “Environmental impact calculator for food”
2. “Food recipe impact calculator”

Seven tools were identified as meeting the first two definitions above, and of these, two were selected for further analysis of how their results are visualized, as they met the final criterion.

2.1.3 Testing the criteria with a recipe

The results obtained from these searches were evaluated based on the definition above, which described an available tool. To ensure that the observed tool meets the third criterion specifically, the following recipe for a Feta Cheese Pasta Dish, inspired by Meny Oppskrifter [20] - a Norwegian grocery store with a website featuring various recipes - was used to calculate the impact for standardization.



Figure 2.1: Baked feta pasta, screenshot from meny.no [20]

Ingredients for Feta Cheese Pasta Recipe for 2 Servings

- 200 grams of pasta
- 250 grams of tomatoes
- 2 cloves of garlic
- 200 grams of feta cheese
- 50ml of olive oil
- 1 teaspoon of salt
- 1 teaspoon of pepper

Ingredients such as dried oregano, lemon peels, and fresh basil were excluded from the recipe for simplicity.

2.2 User feedback survey

With the results obtained from the first methodology described in subsection 2.1, an analysis was conducted based on two topics: the tool's utility and usability. A good prototype of the final website should have the highest interface functionality and design quality, thus a comparison was conducted among the obtained results. This analysis was done by conducting a user feedback survey named "Environmental Impact Calculator Design" as a data collection method. The purpose of this survey was to gather precise information about the user experiences with various environmental impact calculator tools from participants with diverse backgrounds, nationalities, and levels of knowledge regarding environmental impacts. This feedback was used in a co-design

process ¹ to define the standards for the utility and usability of the final product website, ensuring that the tool’s UI and features align with user preferences and expectations.

The survey was created using Google Forms, a free web-based survey administration tool, and was distributed to various audiences by sharing links and QR codes on social media services such as LinkedIn and Facebook Messenger. The responses were collected over 30 days from October 2nd to October 31st, 2024. The full survey can be found in Appendix A.

2.2.1 Survey structure

The survey consisted of five parts. Some open questions allowed for short or long-paragraph answers, while other closed questions required participants to rate on a scale from 1 to 7, using a semantic differential rating system, where the two endpoints represented opposites. All multiple-choice questions included an option “Prefer not to answer” for participants who chose not to respond. Similarly, the open-ended questions were not mandatory, allowing participants to skip them and proceed with the survey without having to respond.

1. Introduction and background

1.1. Description of the survey’s background, purpose, and information on how the project, which the survey is for, is part of the EU-funded RAINFOREST project [10] was presented. The estimated time of how long the survey will take along with the preferred device on which the survey should be filled out was written. Appreciation for participation was also included.

1.2. User’s personal background information (profession, age cohort, gender, nationality, place of residence), user’s prior experience with environmental impact calculators, and familiarity with environmental impacts were asked. No personally identifiable information, such as names or contact details, was collected to ensure anonymity.

2. The user experience and feedback based on the tool’s usability including what the users liked, any challenges they encountered, their assessment of the tool’s interface, the clarity of the results presented, and any suggestions for improvement were asked.

3. Comparison and preferences of evaluated tools identifying any specific features the users found helpful and any suggested features they felt were missing from the tools they would like to see in the new calculator were asked.

4. Questions regarding the interface and features of the results including preferred results’ design format, suggested difficulty in inputting the data, interest in suggestions for sustainable ingredient alternatives, and desired future additions to user interfaces or result displays, were asked.

¹Here, the co-design process refers to a collaborative, user-centered approach that actively involves stakeholders to create solutions tailored to their needs, enhancing usability and experience [21, 22], which will be discussed further next semester.

5. The Open-ended section asked about the regularity of usage of the tool if they had access to the tool with their desired features and other comments or suggestions about the calculators or the survey.

The responses contributed directly to the co-design of the tool's UI and functionality, ensuring that the final design meets user needs and preferences.

2.2.2 Artificial Intelligence (AI)-assisted survey analytics

For analyzing the survey results, ChatGPT, specifically the GPT-4 model, was utilized as an AI-driven tool to extract insights and identify patterns in user feedback efficiently. ChatGPT is a conversational AI model developed by OpenAI that excels at processing large volumes of text to identify recurring themes, categorize information, and summarize data [23], making it useful for qualitative survey analysis. After the survey was closed for more responses, the AI was employed to analyze responses, mainly on what users found valuable and effective, what they disliked or found challenging, and their feedback on result visualization and additional features. It also analyzed user responses regarding their preferences and comparisons among the tools presented in the survey.

However, the use of AI in this context also presented limitations. ChatGPT may oversimplify nuanced feedback, misinterpret ambiguous responses, or fail to capture subtle differences in sentiment [24]. Additionally, its lack of long-term memory may lead to inconsistencies or contradictions in generated content [24]. To address these challenges, all AI-generated results analytics were manually and carefully reviewed and validated against the original survey result to ensure accuracy. Despite these limitations, the use of AI offered several advantages in analyzing the survey results, such as streamlining the analysis process, enhancing the comprehensiveness of data coverage, and together with human oversight, detailed and accurate evaluation of the user feedback survey was conducted.

3 Results

3.1 Evaluation of available environmental impact calculators

Seven tools were identified using a keyword search on Google engine and applying the first two defined criteria for what makes a tool “available”. Each tool was examined to analyze its input data, what the users need to enter, and its results, what the tool calculates and displays. Table 3.1 presents the analysis. Note that these results include general tools that satisfy the first two points of the criteria. This broader evaluation approach provided valuable insights into what was well-executed and what should be avoided in designing the interface of the final prototype. Based on these results, two tools fully met the criteria: the 2nd tool from Table 3.1 - the *Free food carbon footprint calculator*, and the 6th tool from the same Table 3.1 - the *Carbon Calculator*. These two tools allowed users to input ingredients and their weights based on the recipe they want to access for its environmental impact, satisfying the final third criterion, and are highlighted in Table 3.1.

Table 3.1: Overview of the seven tools, including input data and results, with highlighted tools fully meeting the criteria for classification as "available"

Website title	Source	Input data	Results
1 <i>Climate change food calculator</i> (Website link)	[25]	Which food the user would like to know the environmental impact of + How often the user consumes it	Annual GHG emissions from selected food (kg) + Equivalent impacts: distance driven (miles), home heating (days), water usage (eight-minute showers) + Comparison of emissions per serving (kg) across similar food products
2 <i>Free food carbon footprint calculator</i> (Website link)	[26]	Approximate number of servings + Multiple ingredients used + Weight of the ingredients	Carbon footprint per serving (g CO ₂ -eq) + Equivalent impacts: food items (apple, Margherita pizza, and beef burger) consumption, travel distances (petrol car, plane, train in km) + Carbon rating (A to E) + GHG emissions breakdown by ingredient
3 <i>Food carbon emissions calculator</i> (Website link)	[27]	Food category + Food commodity + Set long-distance truck transport + Set quantity purchased by consumer + Set consumer waste percentage	Production, transport, and waste emissions (kg CO ₂ -eq)
4 <i>Quiz: How Does Your Diet Contribute to Climate Change?</i> (Website link)	[28]	Meal/diet type for breakfast, lunch, and dinner + Drinks	Climate impact rating (Low to High) + GHG emission group (%) based on meal choices and protein source preferences

Website title	Source	Input data	Results
5 <i>Foodprint Calculator</i> (Website link)	[29]	Animal and plant-based product consumption frequency (Never to 6+/day) + Optional personal information (Dietary preference, Gender, Age range, Ethnicity, Income, Country)	Annual environmental impact: Carbon (kg), Nitrogen (g), Water (l) + Equivalent impacts: gasoline burned (gallons), polar ice melted (cubic feet), additional carbon (kg), fertilizer pollution (pounds), daily water usage (bathtubs)
6 <i>Carbon Calculator</i> (Website link)	[30]	Approximate number of servings + Multiple ingredients used + Weight and region of origin for the ingredients	Total GHG emissions per serving (g CO ₂ -eq) + Total overall GHG emissions (kg CO ₂ -eq) + Equivalent impact: distance driven (miles)
7 <i>How does what you eat affect the planet?</i> (Website link)	[31]	Country + Number of portions per week for 13 food groups (grains, ..., alcohol) + Option to calculate based on the typical diet of the selected country	Environmental impacts compared to planetary boundaries: Biodiversity loss (species lost/year), GHG emissions (t CO ₂ -eq), Cropland use (hectares), Grazing land use (hectares), Water use (cubic kilometers), Eutrophication (tons of extra nutrients) (from within boundary to greatly exceeds boundary) + Explanation of impacts and further resources + Breakdown of impacts by food group

3.2 Analysis of *Free food carbon footprint calculator* and *Carbon Calculator*

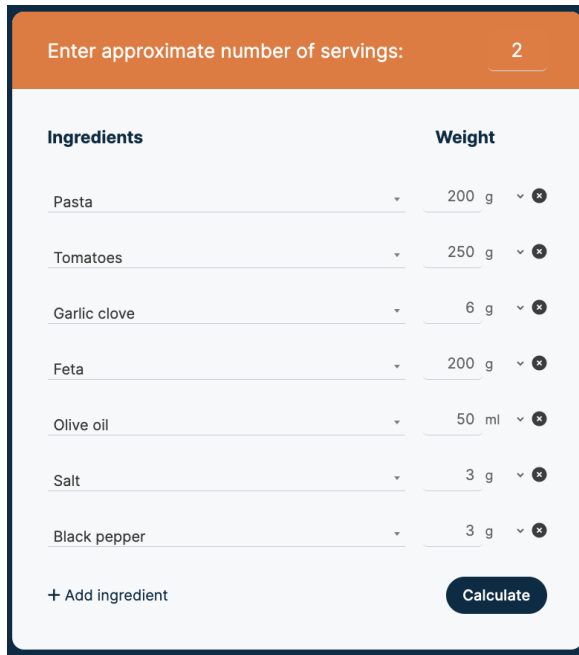
As both the *Free food carbon footprint calculator* and the *Carbon Calculator* met all three criteria for being considered available tools, the Feta Cheese Pasta Dish recipe was used to calculate the recipe’s environmental impacts.

3.2.1 Data entry and input specification

The first tool, the *Free food carbon footprint calculator* from MY EMISSIONS [26], allowed users to manually enter ingredients and their corresponding weights in various units: g, kg, lb, oz, l, ml, tbsp, tsp, and cup. This enabled users to adjust the units based on the most appropriate measurement. When an ingredient was selected, typical serving information was provided, describing the usual portion size of that ingredient. For example, when a user selected pasta from the “Grains, pulses, and legumes” category, the tool suggested a typical serving size *Two handfuls = 75g*. This feature was useful for users who may not know the standard serving size, such as when they are not strictly following a recipe but want to estimate the impact of their meal plan. Ingredients could be added either by typing their name into the search bar or by browsing through the category list to find the appropriate match.

The second tool, the *Carbon Calculator* from The University Caterers Organisation (TUCO) [30], similarly allowed users to add ingredients manually. This was done by first selecting an ingredient category, followed by the specific ingredient. While choosing the food category first was optional, it helped users to browse the various ingredients available within each category. Afterward, users selected the region of origin, with 13 options in total, including three specific UK-based options: *150 miles or over*, *within 150 miles*, and *within 50 miles*. Next, users input the ingredient’s amount, with available units being kg, g, l, and lbs. However, unlike the first calculator, this tool had a limitation when it came to smaller quantities. It did not allow users to input weights below 5g, and values were rounded up to the nearest two-digit multiple of 10. For instance, an ingredient weighing between 1g and 4g was rounded down to 0g, and any weight between 5g and 10g was rounded up to 10g. For example using *Beef: Ground, minced meat from Central America*, which had an emission factor of 23.53kg CO₂-eq/kg according to the tool, the GHG emissions for 5g would be calculated as equivalent to 10g. Ideally, the emissions should have been 0.12kg CO₂-eq for 5g and 0.24kg CO₂-eq for 10g, but this calculator treated them equally. The same applied to liquid measurements, as the calculator translated 1 liter of any liquid as equivalent to 1 kilogram. Even though these roundups only applied to ingredients with smaller weights, they could add up significantly in recipes that include many different ingredients in small portions, especially when using items with high emission factors.

Screenshots of the input data for both calculators, showing all the selected Feta Cheese Pasta Dish recipe ingredients, are presented in Figure 3.1. In the second calculator, the *Carbon Calculator*, the region of origins for all ingredients was set to *unknown* for this calculation, whereas in the first calculator, the *Free food carbon footprint calculator*, there was no option to specify the region of origin. As shown in the screenshots, the *Carbon Calculator* rounded the garlic weight up to 0.01kg, while both salt and pepper were rounded down to 0g.



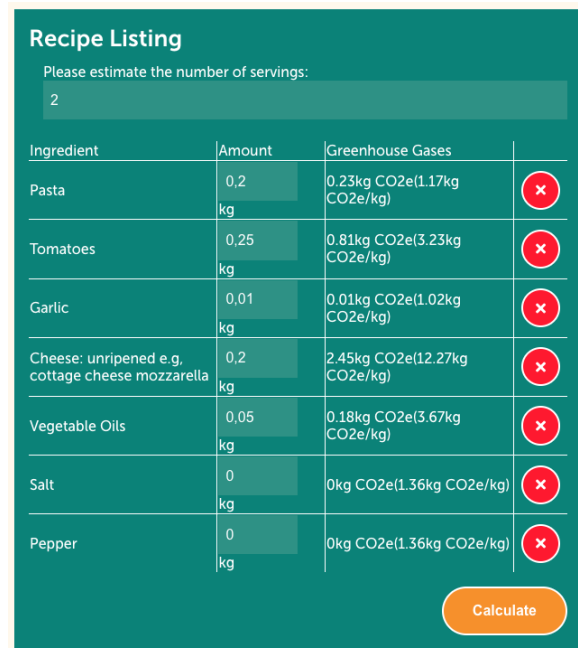
Enter approximate number of servings: 2

Ingredients	Weight
Pasta	200 g
Tomatoes	250 g
Garlic clove	6 g
Feta	200 g
Olive oil	50 ml
Salt	3 g
Black pepper	3 g

+ Add ingredient

Calculate

(a) Free food carbon footprint calculator



Recipe Listing

Please estimate the number of servings: 2

Ingredient	Amount	Greenhouse Gases
Pasta	0,2 kg	0.23kg CO ₂ e(1.17kg CO ₂ e/kg)
Tomatoes	0,25 kg	0.81kg CO ₂ e(3.23kg CO ₂ e/kg)
Garlic	0,01 kg	0.01kg CO ₂ e(1.02kg CO ₂ e/kg)
Cheese: unripened e.g. cottage cheese mozzarella	0,2 kg	2.45kg CO ₂ e(12.27kg CO ₂ e/kg)
Vegetable Oils	0,05 kg	0.18kg CO ₂ e(3.67kg CO ₂ e/kg)
Salt	0 kg	0kg CO ₂ e(1.36kg CO ₂ e/kg)
Pepper	0 kg	0kg CO ₂ e(1.36kg CO ₂ e/kg)

Calculate

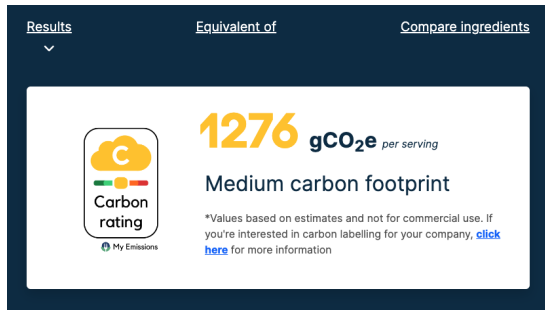
(b) Carbon Calculator

Figure 3.1: Ingredient selection processes for the Feta Cheese Pasta Dish recipe on two calculators, screenshots from the corresponding websites [26][30]

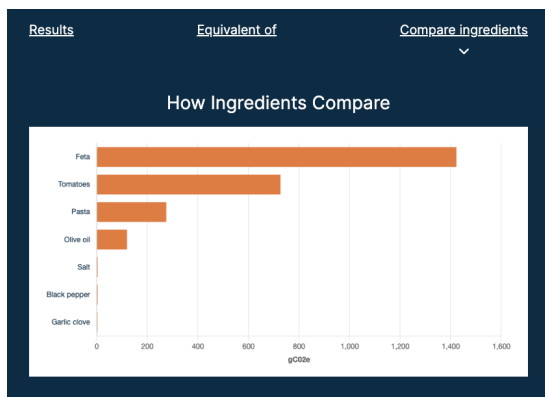
3.2.2 Result visualization and interpretation

Once users have entered the input data and calculated the impact, the first tool, the *Free food carbon footprint calculator*, presented the results in three formats; carbon footprint per serving (in g CO₂-eq), equivalent impacts, and an emission breakdown by ingredients. Screenshots of these results are shown in Figure 3.2. As seen in Figure 3.2a, the first part of the result - carbon footprint per serving - was displayed in g CO₂-eq, along with a carbon rating defined by the tool. According to MY EMISSION [26], this Carbon rating, ranging from A (Very Low) to E (Very High), was based on the recipe's carbon intensity, meaning the emissions produced for every 1kg of food. This rating system was inspired by existing nutritional leveling and aligned with the latest climate science, as stated by the website's creators. The color-coding of the emission numbers and the rating system made it easier for the users to quickly identify where their recipe falls on the scale. The results were then translated into equivalent impacts, as shown in Figure 3.2c. Since raw carbon emission data were harder to interpret, converting these figures into more familiar contexts helps users better understand the impact. This section was divided into two parts: the first showed the equivalent impact of eating different foods, such as apples, slices of Margherita pizza, and beef burgers. The second part illustrated the equivalent impact of various modes of transportation (e.g., driving a petrol car, flying, or taking a train) if the user were to eat the selected recipe once a week for a year. Finally, the last part of the results, shown in Figure 3.2b, presented a breakdown of emissions by ingredients using a horizontal bar graph. The ingredient names were listed on the left (y-axis) and their corresponding emissions on the bottom (x-axis), using the same unit of measurement as the carbon footprint results from the first part.

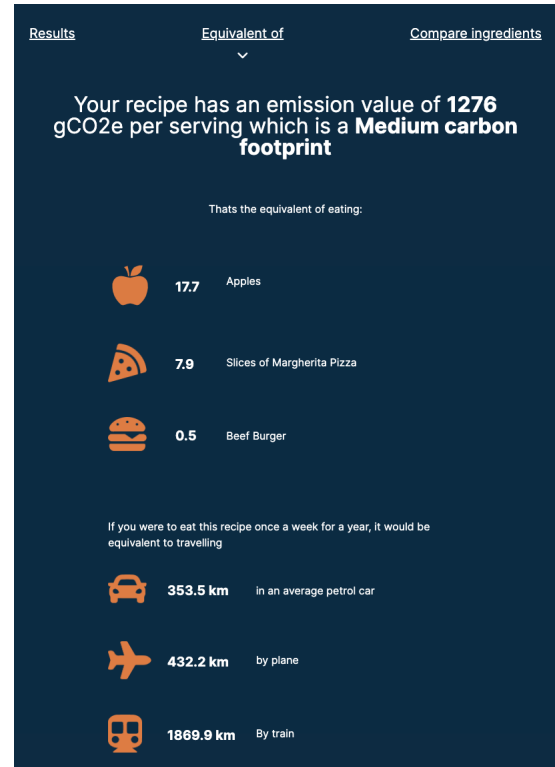
In summary, the *Free food carbon footprint calculator* allowed users to calculate the carbon footprint per serving of a dish, provided equivalent impact comparisons, and offered a visual representation of the emission breakdown by ingredients.



(a) Total carbon footprint per serving



(b) Equivalent impacts



(c) Emission breakdown by ingredients

Figure 3.2: The Free food carbon footprint calculator results, screenshots from the website [26]

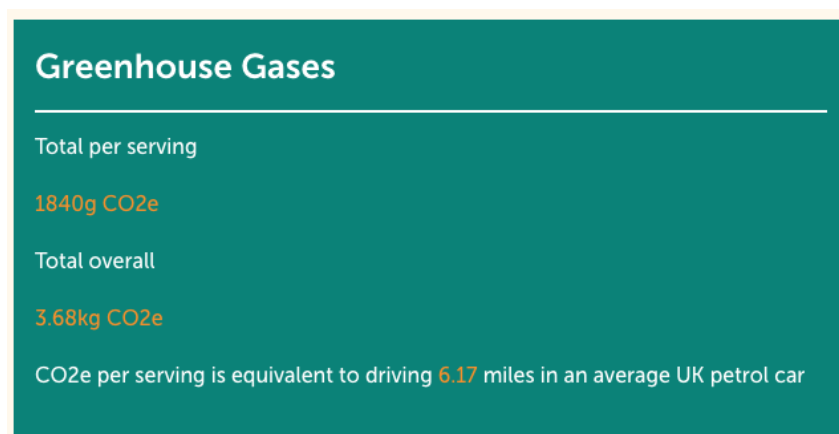


Figure 3.3: The Carbon Calculator result, screenshot from the website [30]

The second tool, the *Carbon Calculator*, provided a simpler result than the first tool. It presented the total GHG emissions per serving in g CO₂-eq and the overall emissions in kg CO₂-eq. As an equivalent impact, it showed the distance (in miles) an average UK petrol car could drive based on these emissions. A screenshot of the calculated result is shown in Figure 3.3 above.

3.3 User feedback survey result

In the user feedback survey, participants were asked to evaluate a total of three of the calculators analyzed above: the *Free food carbon footprint calculator*, the *Carbon Calculator*, and the WWF’s “*How does what you eat affect the planet?*” tool, respectively as the tool A, tool B, and tool C. Although the last tool did not meet the final criterion for tool availability, it stood out by presenting results across various environmental impacts, not just GHG emissions, but also biodiversity loss, cropland use, and other factors, based on a customized diet profile, so it was included in the survey. The survey structure defined in 2.2.1 was used with the mentioned three tools presented. When describing the results obtained from the survey, the terms “users” and “participants” were used interchangeably.

3.3.1 Participants demographics and background information

A total of 36 participants answered the survey, with approximately 42% indicating that they work in the private sector or industry, 28% in the university or academic sector, and the rest identified as students. A majority of participants, approximately 64%, were between 21 and 30 years old, followed by 24% between 31 and 40 years old. Additionally, a large portion of the participants were male, also around 64% (Figure 3.4).

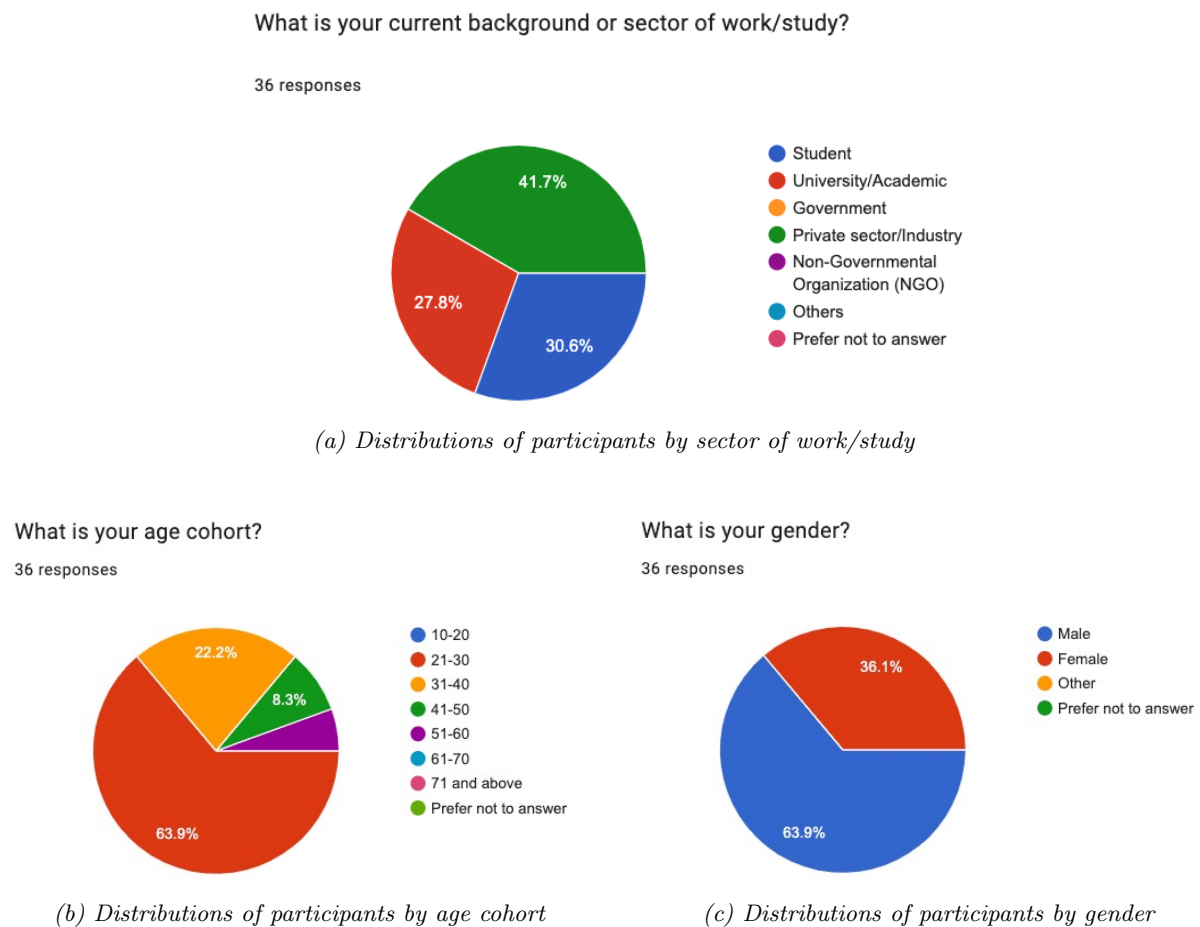
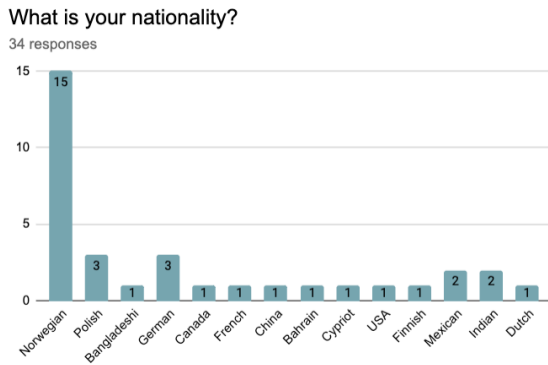
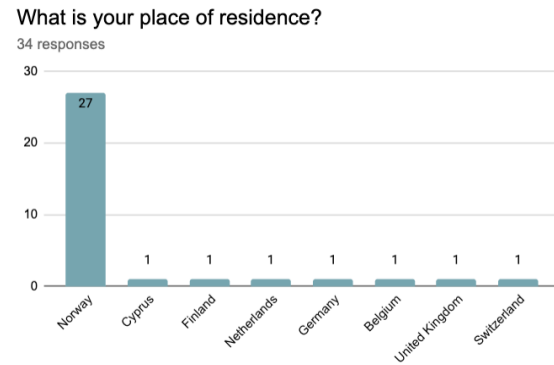


Figure 3.4: Distributions of participant demographics by sector, age, and gender, screenshots of the Google survey results

The participants came from diverse countries and places of residence. Out of the 36 participants, 34 shared their nationality and current location. The survey received responses from participants of 14 different nationalities, with about 44% identifying as Norwegian, making them the largest national group. Altogether, participants resided in 8 different locations, with 27 out of 34 respondents, roughly 80%, currently living in Norway (Figure 3.5).



(a) Distributions of participants by nationality

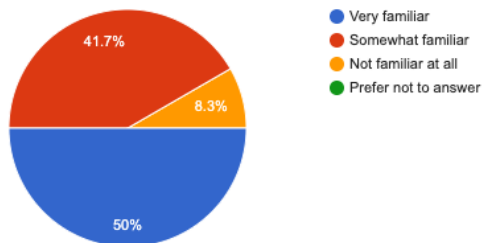


(b) Distributions of participants by place of residence

Figure 3.5: Participant demographics by nationality and place of residence, screenshots of the Google survey results

How familiar are you with various environmental impacts, such as greenhouse gas emissions, land use, eutrophication, etc.?

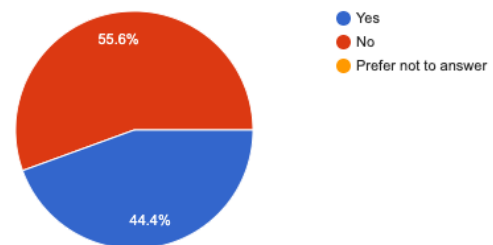
36 responses



(a) Familiarity with environmental impacts

Have you used any environmental impact calculators before?

36 responses



(b) Experience with environmental impact calculators

Figure 3.6: Participant familiarity and prior experience with environmental impact calculators, screenshots of the Google survey results

Regarding participants' familiarity with environmental impacts and prior experience with environmental impact calculators, half were already very familiar with various environmental impacts, while about 42% answered being somewhat familiar (Figure 3.6). However, slightly over half of the participants had not previously used environmental impact calculators (Figure 3.6). Reasons for this familiarity with various environmental impacts were likely due to the survey's distribution paths and audience types. The survey was shared among the author's classmates and study colleagues, as well as on LinkedIn, targeting an audience largely involved in academia. Additionally, sharing the survey on Nordic Semiconductor's employee forum introduced participants from the private sector, influencing results on participants' backgrounds.

3.3.2 Technical issues and their impact on survey results

On October 9th, 2024, tool A presented in the survey encountered an error that prevented users from manually adding ingredients, making the tool unusable. The website's creator was contacted via email on the same day to address the issue. The website resumed normal functioning on October 11th. This problem affected three participants who completed the survey on October 9th and October 10th.

On October 14th, 2024, tool B from the survey experienced an SSL (Secure Sockets Layer) error, displaying the error message "Your connection is not private". This issue prevented most participants from accessing tool B that day, though one could still access it. Since the participants could technically proceed by accepting the expiration of the tool's security certificate, the survey description was edited clearly so that, tool B remained accessible although not secure despite the server issues. Participants were informed that they could continue to access the tool at their own risk, with the understanding that their information might not be protected. This technical issue affected a total of three participants.

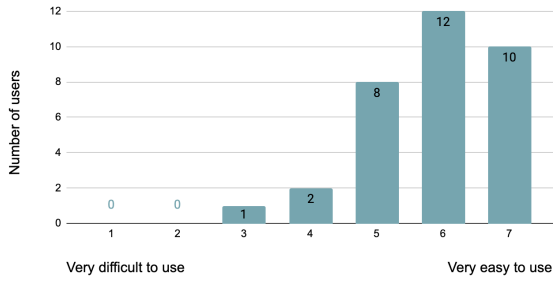
3.3.3 User-friendliness and usefulness of the three tools

Due to the technical issues mentioned, three responses were affected for each of tools A and B. The three participants who could not access tool A rated its user-friendliness and usefulness relatively low, giving it an average score of 3.7 out of 7 for the tool's user-friendliness and 1.3 for usefulness. A similar situation occurred with tool B, where another three participants were affected and provided an average score of 3 for both the tool's user-friendliness and usefulness.

The unaffected results for user-friendliness and usefulness of all three tools are shown in Figure 3.7. Tool C had no technical issues during the survey period, so its results reflect the full set of responses from all 36 participants. The complete results, including participants' responses affected by technical issues, for user-friendliness and usefulness of all three tools are presented in Appendix B. Based on these results, the averages for each of the tool's user-friendliness and usefulness were calculated, along with their standard deviations (Table 3.3). Tool A scored the highest average value for user-friendliness along with the smallest standard deviation compared to the other tools. On the other hand, tool C received the highest average score for usefulness with the second-smallest standard deviation, indicating that users generally had similar perceptions of its usefulness, though there was slightly more variation in their responses compared to Tool B.

How user-friendly did you find the interface of tool A?

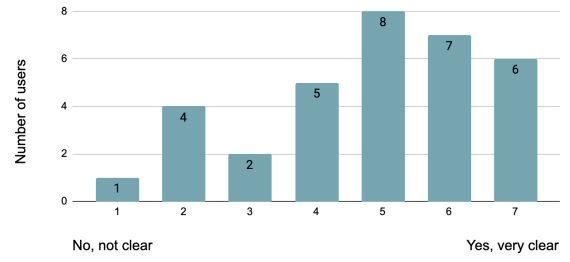
33 responses



(a) Tool A's user-friendliness results

Did the tool A provide clear and useful results on the environmental impact of food?

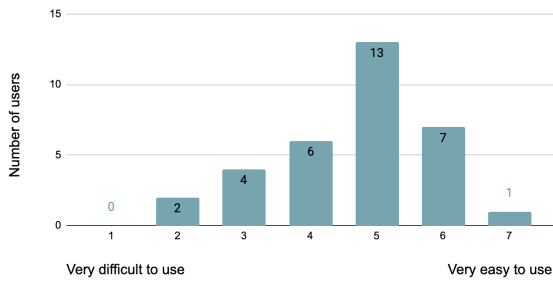
33 responses



(b) Tool A's usefulness results

How user-friendly did you find the interface of tool B?

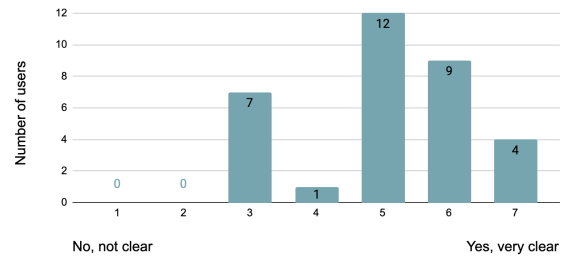
33 responses



(c) Tool B's user-friendliness results

Did the tool B provide clear and useful results on the environmental impact of food?

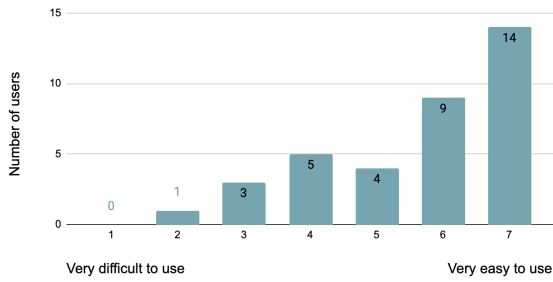
33 responses



(d) Tool B's usefulness results

How user-friendly did you find the interface of tool C?

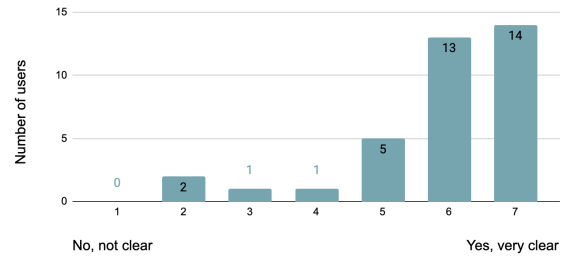
36 responses



(e) Tool C's user-friendliness results

Did the tool C provide clear and useful results on the environmental impact of food?

36 responses



(f) Tool C's usefulness results

Figure 3.7: Unaffected results for each of the tool's user-friendliness and usefulness, screenshots of the Google survey results

Table 3.3: Average user-friendliness and usefulness of tools A, B, and C with standard deviations, with the highest average values highlighted

Tool	User-friendliness		Usefulness	
	Average	Standard deviation	Average	Standard Deviation
Tool A	5.85	1.03	4.82	2.14
Tool B	4.67	1.22	5.06	1.30
Tool C	5.64	1.48	5.89	1.35

3.3.4 User experience and feedback results

Participants were asked to evaluate each tool in depth, focusing on usability aspects such as what they liked, disliked, or found challenging and their impressions of how results were visualized. In addition to scoring each tool's user-friendliness and usefulness, presented in 3.3.3, they provided open-ended responses for their feedback. This subsection explores the written feedback on user experience for each tool, analyzing and summarizing key themes and common points from the responses. A complete set of survey responses is provided in Appendix C.

Tool A

Based on the results from 36 participants, three responses were affected by technical issues and two blank responses were excluded. As a result, 31 responses were analyzed regarding what participants found valuable and effective in tool A, while 27 responses were analyzed for feedback on dislikes and challenges with tool A. Additionally, 29 responses were analyzed on how tool A displayed results, and 15 responses were reviewed for additional feedback.

What users found valuable and effective

1. Ease of use

Many participants appreciated the tool's ease of use, describing it as "intuitive", "straightforward", and "efficient". Interactive elements such as dropdown searches and the option to add multiple ingredients were particularly valued, and the simplicity and efficiency of the tool were highlighted as positive aspects.

2. Typical serving size suggestions

Participants found that including typical serving sizes and weight approximations for ingredients helped estimate portions and assess serving sizes accurately.

3. Ingredient options and measurement flexibility

The comprehensive list of ingredients and the flexibility to choose various food items in different units of measurement received positive feedback from users.

4. Comparison features

Participants appreciated the tool's comparison features, which helped contextualize results and made them more relatable.

What users disliked and found challenging

1. Limited scope of environmental impacts

Some users expressed that the tool focused exclusively on CO₂ emissions and did not consider other environmental impacts like land and water use.

2. Difficulty with ingredient entry and navigation

The process of entering multiple ingredients for complex recipes was viewed as time-consuming and tedious. Some users found the ingredient list long and sometimes difficult to navigate, suggesting the inclusion of pre-made meal options, improved categorization,

a more intuitive organization of ingredients, and a more visible search function.

3. Lack of ingredient origin and data transparency

The tool lacked information on ingredient origin, seasonality, and production methods including transportation, which users felt limited their understanding of the recipe's environmental impact. Additionally, they wanted more transparency about how calculations were made, including data sources and assumptions behind the results.

4. Confusing metrics and comparison

The metric used for the results, "g CO₂-eq per serving" was unclear to several users, with some reporting that they did not understand what "serving" meant. Without context or a clear definition provided, they struggled to understand what this metric actually represented. The phrase "per serving" added further ambiguity, as users were unsure how "serving" was defined or how it corresponded to real-world portion size.

In summary, tool A was praised for its simplicity and helpful features, scoring the highest user-friendliness among the three tools (Table 3.3), but users felt it could be significantly improved by expanding its environmental metrics, enhancing ingredient entry and navigation, and providing clear data explanations and ingredient information.

Positive feedback on tool A's result visualization and additional feedback

1. Clear and simple display

Users found the results clear and easy to read, making it straightforward for them to understand the information presented which was also visually pleasant.

2. Equivalent comparisons

Users appreciated the ability to compare their carbon footprint to other food types and activities, such as travel distances, helping them contextualize the impact of their food choices more relatable.

3. Carbon rating scale

Users found the Carbon rating, ranging from A (Very Low) to E (Very High), helpful for quickly understanding the significance of their results.

4. Ingredient breakdown

The ability to view a breakdown of carbon impact by ingredients was particularly appreciated to see which ingredient contributed the most to the overall impact. Some users wanted more disaggregated results for deeper insights.

Users generally found the presentation and visualization of results in Tool A to be clear, visually appealing, and easy to understand through several contextualization elements such as equivalent comparisons, a clear Carbon rating scale, and an ingredient breakdown of the overall carbon impact.

Tool B

Out of 36 participants, 2 responses were affected by technical issues and 5 were blank, so they were excluded. Consequently, 29 responses were analyzed to identify what participants found valuable and effective in Tool B, and another 29 responses were examined for feedback on aspects they disliked or found challenging. Then 25 responses were reviewed regarding how tool B displayed its results, and 20 responses were considered for additional feedback.

What users found valuable and effective

1. Region of origin feature

Nearly half of the participants (13 out of 29) gave positive feedback on tool B's ability to show the region of origin for input ingredients. This feature gave them a more detailed understanding of the environmental impact specific to each food's origin.

2. Relatable comparison

Some users appreciated the tool's feature of comparing carbon emissions to real-life activities, such as car driving, which made the impact more relatable and easier to understand.

3. Simplicity of the tool

The tool was described as intuitive, easy to use, and simple, with users noting the effectiveness of the one-screen UI design. Step-by-step instructions on how to use the tool and clear sorting categories for available ingredients helped the tool's simplicity and ease of use.

What users disliked and found challenging

1. UI and design critique

Tool B's design received criticism for not feeling clean or visually appealing, with some users finding it too colorful or cluttered. The lack of clear divisions made it challenging to see different sections clearly, and some users noted that too many clicks were required rather than allowing a simple search feature. Although this search function was available, it was not visible enough for users to notice easily.

2. Ingredient availability

Some users felt that the tool had a limited selection of ingredients and did not include certain items that they expected to find.

3. Lack of comparative and contextual impact information

Although users appreciated comparing the results to car driving, some wanted a broader context, such as comparisons with other travel methods or an indicator of "high" or "low" carbon footprint rating. Additionally, the tool's UK-centric focus limited its relevance for some users.

In summary, Tool B was valued for its regional origin feature, relatable comparisons to real-life activities, and user-friendly design, though it received mixed feedback on its visual layout.

Users suggested improvements in ingredient availability, comparative metrics, and clearer contextualization of impact results to make the tool more broadly applicable and informative.

Positive feedback on tool B's result visualization and additional feedback

1. Relatable comparison to car driving

Users consistently appreciated the comparison of their meal's carbon emissions to a familiar everyday activity, car driving. This feature was frequently mentioned as it made the results more relatable and meaningful.

2. Clear per serving and total impact display

Some users valued the clearly distinguished results between total per serving impact and total overall impact, which made the information easier to interpret.

Overall, tool B was praised for its in-depth features, such as allowing users to select the region of origin for ingredients and providing practical comparisons of results in driving distance by car. However, it would benefit from a broader ingredient database, a refined UI, and expanded result comparisons.

Tool C

Based on the results from 36 participants, 2 blank responses were excluded. Since there were no technical issues with this tool, a total of 34 responses were analyzed regarding what participants found valuable and effective in Tool C, while 29 responses were examined for feedback on dislikes and challenges. Additionally, 32 responses were reviewed on how Tool C displayed results, and 16 responses were considered for additional feedback. Except for the additional comment question, Tool C received the most feedback from participants compared to the other tools.

What users found valuable and effective

1. Country-specific diet information

Many participants appreciated the tool's suggestions on the typical diet presets for various countries, especially their own. This feature allowed them to compare their diet with national averages and with typical diets from other countries, giving them context on their consumption patterns and their environmental impacts.

2. Range of environmental impact metrics

The tool included detailed insights into the diet's environmental impacts and health metrics, such as daily caloric intake and whether the diet was optimal for health. In addition to GHG emissions, the tool provided five other environmental impact metrics including comparisons to planetary boundaries, which received positive feedback from users.

3. UI and interactivity

The tool's intuitive design and interactive sliders received widespread praise from participants. They appreciated real-time updates as they adjusted diet choices along with easily modifiable portion settings without needing specific product weights. The

immediate feedback on the same screen helped users quickly understand the consequences of their food choices.

What users disliked and found challenging

1. Complexity and information overload

Some users had mixed reactions to the tool's functionality, with feedback indicating it felt initially confusing, overwhelming, hard to use, and cluttered. They described the tool as "noisy", with too much information displayed at once, which required more time to understand both how to use the tool and interpret the visualized results.

2. Portion-based measurements

Although the ease of adjusting portions across different food categories was praised, some users criticized the reliance on portion-based rather than weight-based measurements, finding it difficult to accurately interpret portion sizes (e.g., "30 portions per week of fats & oils").

3. Difficulty in personalization

Some users noted that tool C lacked adaptability for personal characteristics, such as specific body morphology, activity level, and gender. They felt that the tool's health recommendations were overly general and sometimes promoted very low or unsustainable caloric intake as ideal and environmentally sustainable. Also, some users missed the detailed regional options for food types available in other tools.

Tool C's detailed and interactive features were well-received, with users appreciating its thorough impact categories, ability to compare diets by country, and instant feedback on adjustment on the same screen. However, some found it initially overwhelming, and the use of portions instead of weight units was confusing. Limited options for personal customization and a lack of regional features along with a cluttered layout also posed challenges, though many still valued its insightful view of diet impacts, scoring the highest usefulness among the three tools (Table 3.3).

Positive feedback on tool C's result visualization and additional feedback

1. Comprehensive breakdown

Users appreciated the detailed breakdown of multiple environmental and health categories, each of which could be clicked on for further information. This feature helped users better understand the full impact of their diet on both personal and planetary health. The food category breakdown for each impact result was also well-received.

2. Result visualization

The clear visuals, threshold indicators, live-updating feedback, and interactive elements such as the thumbs-up symbol and color indications made it easier for users to stay motivated in meeting environmental goals. Suggestions on how to adjust their diets further enhanced their motivation.

Tool C was praised for its wider range of environmental impact categories, providing users with insightful information into both health and planetary health based on their diet style. It stood

out by scoring the highest usefulness, particularly because users appreciated the interactiveness and graphic representation of the results. Feedback suggested improvements for clearer UI and the ability to regionalize the food types and personalize physical profiles.

3.3.5 Comparison and preferences

For the third part of the user feedback survey, participants were asked to choose which of the three tools they preferred the most along with the reasoning behind and features in any of the tools they found helpful and finally, any suggested features they felt were missing and would like to see in the new calculator that this project aims to create the criteria of.

To be consistent with other results presented above, six responses affected by technical issues were excluded for analyzing the results regarding tool preferences and their associated reasoning. However, these responses were included when analyzing feedback on helpful features and suggested improvements, as the technical issues did not influence these aspects. The complete results of this third part of the survey can be shown in Appendix D

Over a majority of the participants, approximately 63%, found tool C the most preferred tool (Figure 3.8).

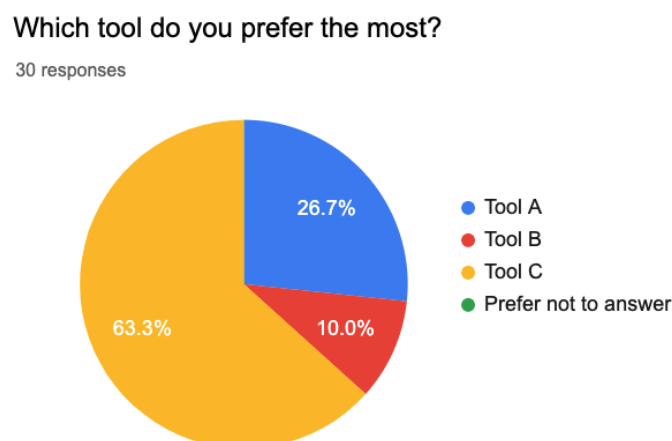


Figure 3.8: Distributions of participants' most preferred tool, screenshot of the Google survey results

Reasoning for preferred tool selection

1. Tool A - Free food carbon footprint calculator

Users who preferred tool A highlighted its simplicity and ease of use as the main reasons for their choice. The inclusion of helpful features like a search field and guided support during data entry were appreciated. One respondent noted the tool's focus on household-level purchase impact, while another suggested that combining its strength with tool C's broader dietary analysis could give even better results for calculating the impact of an entire diet compared to a single meal.

2. Tool B - Carbon Calculator

The only reasoning provided for tool B being the most preferred was that it was easy to

use and offered regional-level results.

3. Tool C - *“How does what you eat affect the planet?”*

Tool C was chosen as the most preferred by the majority of users, who emphasized its ability to provide comprehensive insights and its user-friendly interface. The calculator’s results, presented across multiple environmental impact categories, including planetary boundaries and health aspects, were highly appreciated. Users also highlighted the interactive features, such as the sliding scale and several participants praised the tool’s result presentation and visualization, noting that it offered a more complete, holistic view of their diets.

The analysis of the features users found helpful across the tools showed a strong preference for engaging, user-friendly design elements and the contextualization of results with minimal complexity. As mentioned in the user experience and feedback results, suggested serving sizes and various units provided by tool A were praised.

For the features users would like to see in the new tool, many expressed a desire for an enhanced user experience, such as AI integration to suggest ingredients based on recipes, the ability to save and track progress, and an app that allows users to track their diet over time. Additionally, there was also a desire for broader impact considerations beyond climate change, as well as the ability to trace the impacts along supply chains. Along with two other questions focused on gathering user feedback to enhance the functionality, design, and overall usability of environmental impact calculators, a more detailed analysis of desired features for the future tool is presented in 3.3.8 Desired features.

3.3.6 Interface and features

After evaluating and comparing tools A, B, and C, users were asked to assess the interface and result features, such as the design format. They were allowed to select multiple preferences for how results should be displayed, including visual charts and graphs, simple numbers and percentages, and comparisons to real-world activities. The most preferred option, selected by 34 out of 36 respondents, was visual charts and graphs, aligning with earlier feedback that praised the use of visual representations in the tool. The second most favored method was comparisons to real-world activities, which received 26 votes. Lastly, 11 participants preferred simple numbers and percentages. These preferences are summarized below (Figure 3.9).

Would you prefer the results to be displayed as:

36 responses

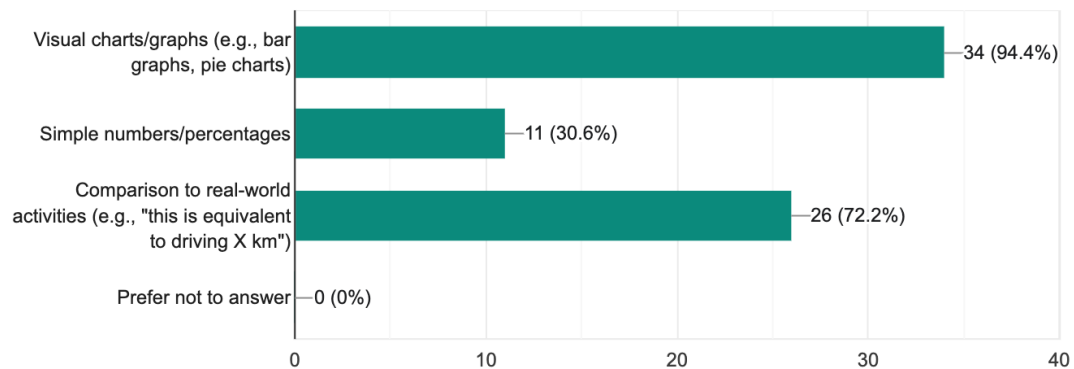
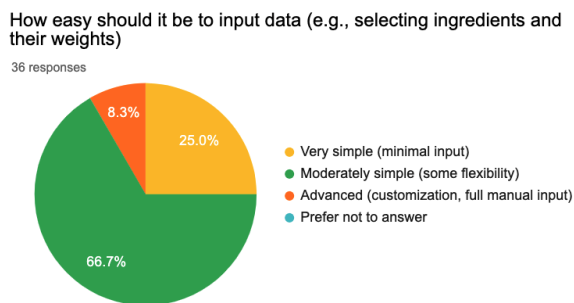
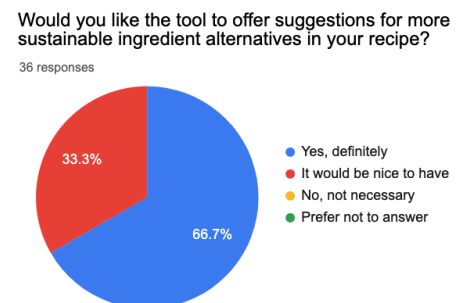


Figure 3.9: Result visualization preferences, screenshot of the Google survey results

When asked about the desired ease of data input, more than half of the participants, 24 out of 36, preferred a moderately simple approach with some flexibility, particularly for tasks like selecting ingredients and specifying their weights. Then nine participants, equaling 25%, favored a very simple input process with minimal input. These results highlight the users' preference for a balance between simplicity and flexibility. Regarding the inclusion of suggestions for more sustainable ingredient alternatives, the idea received strong support, with 24 out of 36 participants answering "Yes, definitely". This reflects a clear interest in tools that promote sustainability and detailed results are presented below (Figure 3.10).



(a) Preferred ease of data input



(b) Sustainable ingredient alternatives

Figure 3.10: Input preference and interest in sustainable ingredients suggestion, screenshots of the Google survey results

Twelve responses were received regarding feedback on user interface elements or result presentations that users felt were missing and should be incorporated into the new tool. The complete list of these responses is presented in Appendix E. Many participants expressed a strong interest in enhancing the tool's visual and interactive features. Building on the graphical elements of Tool C, users suggested incorporating real-world comparisons, images, and emoticons to improve clarity and engagement. As this question was also aimed at gathering user feedback to identify missing elements, whether related to features, user interface components, or result

presentations to improve the future impact calculator, a more detailed analysis of desired features for the future tool is provided in 3.3.8 Desired features.

3.3.7 Open-ended section

Finally, for the final section of the survey, participants were asked to share how regularly they would use the tool if it included their desired features, along with any additional comments or suggestions regarding the calculators or the survey itself.

Half of the participants answered that they would use the tool occasionally, followed by a significant portion choosing monthly, and then rarely (Figure 3.11).

If a calculator with your desired features were available, how often would you use it?

36 responses

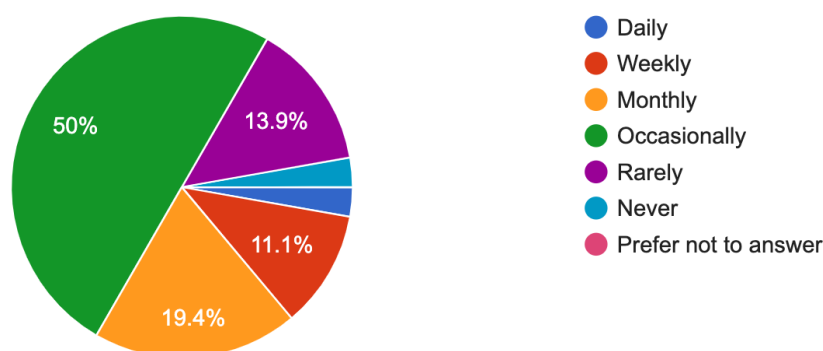


Figure 3.11: Expected frequency of tool usage, screenshot of the Google survey results

This first question particularly was asked to gain insights into the potential practical application of the tool and highlight user priorities for future improvements. The final question received a mix of positive feedback and suggestions for improving the tool, and the complete results are presented in Appendix F. Participants appreciated the thoroughness of the survey and the project's objectives, with several offering constructive ideas for enhancing the calculators. The analysis of these results is presented in the following 3.3.8 Desired features section.

3.3.8 Desired features

In three sections of the survey - Comparison and preferences, Interface and features, and Open-ended section - participants were asked three specific questions, listed below, to identify their preferences, desired features, and gaps in existing environmental impact tools.

1. What feature(s) were missing from any of the tools that you would like to see in a new environmental impact calculator?
2. What user interfaces or result presentations do you feel are missing and would like to see in the future?

3. Do you have any other comments or suggestions about the calculators or the survey?

While the results of these questions were briefly analyzed in their respective sections above, this subsection provides a more in-depth analysis of the features participants thought were essential for a future environmental impact tool for food recipes and the functionalities they felt were missing. The responses received for the three questions were categorized into key features that the users found as missing or needing enhancement.

Interactive features and user engagement

Many users appreciated the interactive features that allow them to add ingredients and edit ingredients, seeing how these changes impact the environmental footprint in real time. They also suggested more visual elements, such as the ability to drag and drop food ingredients onto a plate to calculate their environmental impacts. Users also wanted the option to modify existing recipes to see how different ingredients particularly plant-based protein alternatives replacing meat, affect the overall impact. Finally, there was a strong desire for the tool to suggest more environmentally sustainable ingredient alternatives and allow for side-by-side comparisons of these options.

Result visualization and explanation

Users desired for the results to be presented with more than just raw numbers and texts. They preferred engaging visuals, such as an interactive tool that would allow them to click and learn more about the environmental impacts like eutrophication, acidification, and land use, that are often unfamiliar to those without a sustainability background. Real-world comparisons, such as relating the impact results to daily activities, helped users to understand the results better. Several participants also requested visual aids, such as ingredient icons, to make the tool more engaging and help identify food items. Users found the detailed and disaggregated impact data for different ingredients in tool C particularly useful, as it highlighted the “hotspots” in their food choice, showing which ingredients contributed most to the environmental impact. Many expressed a desire for similar features in the future tool.

Sustainability focus

Some users noted the importance of providing recommendations or limits for certain ingredient usage to encourage a more sustainable recipe style. They suggested features that could guide users on how frequently they should consume specific recipes or ingredients with high environmental impacts. This type of feature, combined with the tool’s ability to suggest more environmentally sustainable ingredient alternatives, would help users understand not only the direct environmental impact of their food choices but also how adjusting their habits can contribute to sustainability goals over time.

Accessibility and integration features

Participants suggested several features to enhance the tool’s accessibility, such as integrating it into online recipe websites or developing a weekly grocery impact calculator. They also proposed creating a mobile app or embedding the tool in shopping apps for real time insights

during grocery shopping progress. Also, AI-driven functionalities like predicting or auto-filling ingredients based on a dish name were highlighted as ways to improve usability and efficiency.

Based on the 41 written responses regarding desired features, excluding irrelevant comments unrelated to features, Table 3.4 provides an overview of the specific features mentioned or requested by participants the most, along with the number of mentions for each.

Table 3.4: Most desired features by participants

Feature	Number of Mentions
Visual and interactive design (e.g., drag-and-drop, ingredient icons, graphs)	8
Sustainable ingredient recommendations and frequency-of-use suggestions	8
Integration into existing tools (recipe websites, shopping apps, mobile apps)	5
Editable and pre-calculated meal features	3

These insights highlight the key features that should be prioritized in the design of the future environmental impact calculator for food recipes, which will be developed next semester.

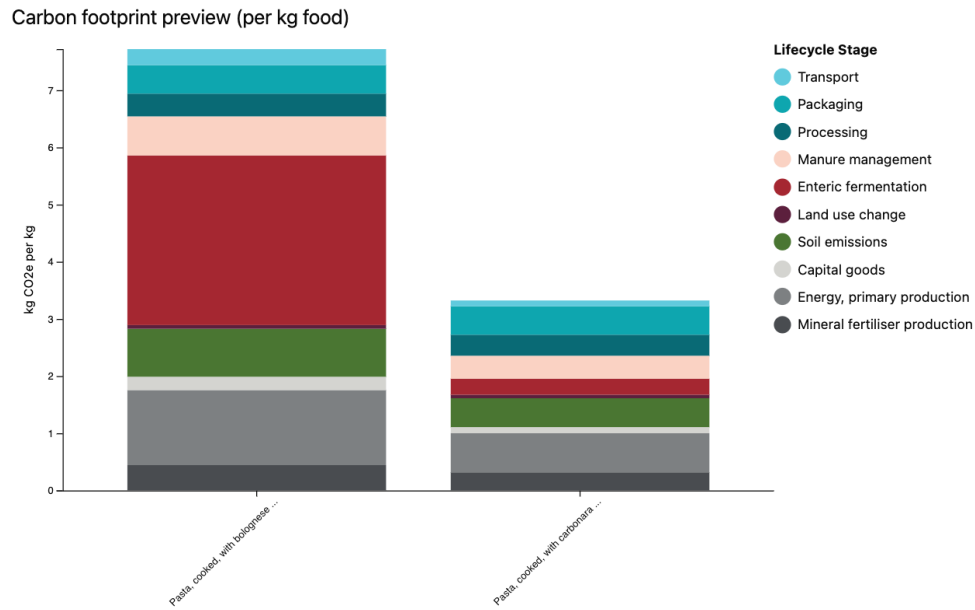
3.4 Analysis of SAFAD - Sustainability Assessment of Foods and Diets

On November 12th, 2024, after the user feedback survey was completed, an additional tool was discovered, the “SAFAD - Sustainability Assessment of Foods and Diets”. This tool calculated the environmental impact of foods and diets using eight environmental indicators and indicators for animal welfare and antibiotic use [32]. Even though the SAFAD tool did not fully meet all three criteria for being classified as “available” as defined earlier, it still offered a great variety of pre-made food dishes with multiple ingredient variations. Additionally, the environmental impact results it provided covered a wide range of impacts, not just carbon footprint. Since the user feedback survey had already been closed, user feedback for this tool was not obtained. However, this tool was along with the previously evaluated calculators in the discussion chapter to define standards for the final prototype’s utility and usability.

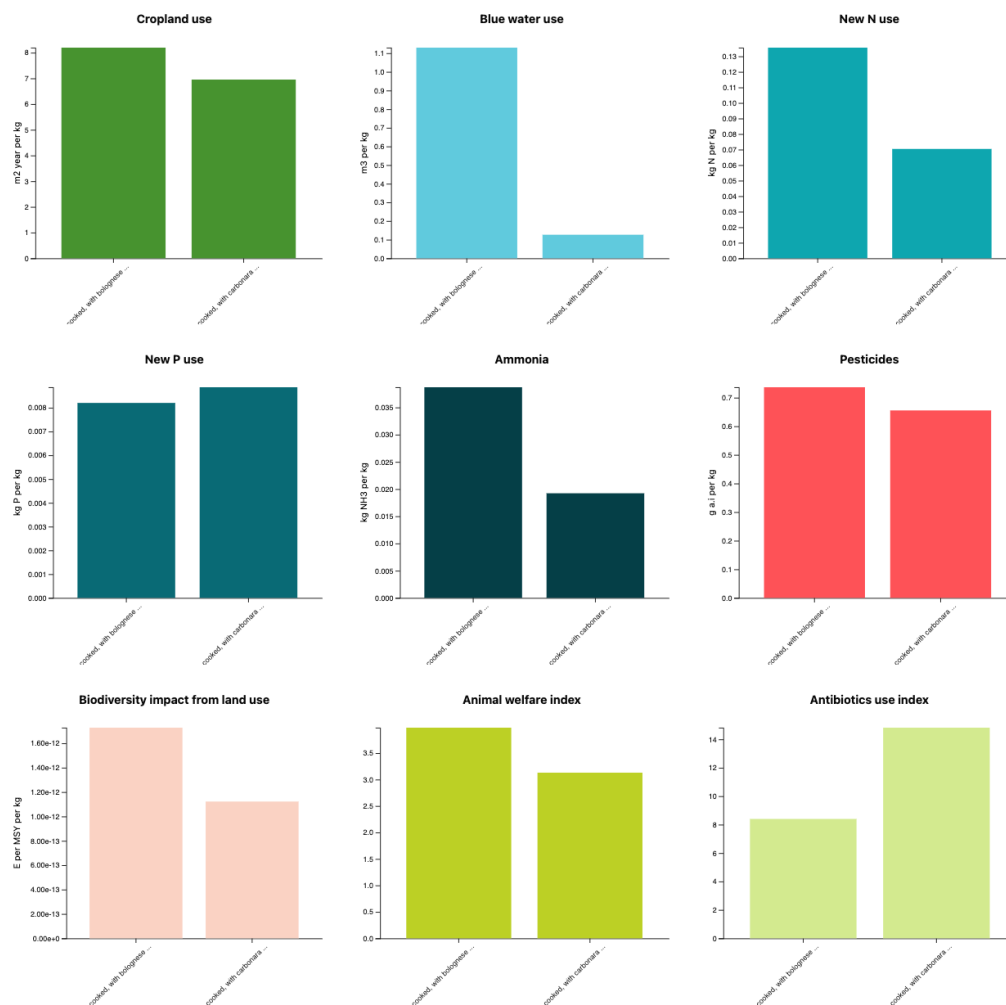
The SAFAD tool first allowed users to select from a variety of finished foods and ingredients, enabling comparisons of individual ingredients and/or dishes. For example, users could customize a pizza by selecting specific ingredients such as spinach or tomato for vegetables, or ham or chicken for proteins. However, unlike what the future tool will offer, SAFAD did not allow users to combine various ingredients to formulate a complete dish.

The first part of the results displayed the food names along the x-axis (bottom) and the corresponding carbon footprints along the y-axis (left), measured in kilograms of CO₂-eq per kilogram of food under the Carbon footprint preview section. These carbon footprints were further broken down into 10 distinct life cycle stages, including transport, packaging, processing, soil emissions, and energy use. The second part of the results featured nine graphs under the Environmental Impacts section. These graphs covered key environmental metrics such as Cropland use, Bluewater use, New Nitrogen (N) use, New Phosphorus (P) use, Ammonia emissions, Pesticides, and Biodiversity impact from land use, as well as two separate graphs for the Animal welfare index and Antibiotics use index. After presenting the environmental impacts of individual foods, the tool then displayed the impacts of the selected diet. These results included key metrics such as the diet’s alignment with planetary boundaries according to the website’s selected diet based on a scientific article, contributions to various environmental impacts from different food groups and raw commodities, and the breakdown of the carbon footprint by various gases and life cycle stages. These diet-related results were influenced by the country chosen by the user. The tool allowed users to select from nine countries, with Sweden set as the default. Note that the environmental impacts of the same food varied depending on the country selected, as the carbon footprint and other environmental metrics were adjusted based on local production, transportation, and processing conditions. However, not all countries had the diet impact results available. For instance, data for Poland was not included in the results.

The following (Figure 3.12) presents the first two sections of the results for two selected pasta dishes as examples: cooked pasta with bolognese sauce (shown in the first column) and cooked pasta with carbonara sauce (shown in the second column). These results are based on Sweden as the selected country option.



(a) Carbon footprint preview of pasta dishes: Bolognese vs. Carbonara Sauce (Sweden selected)



(b) Environmental impacts of pasta dishes: Bolognese vs. Carbonara Sauce (Sweden selected)

Figure 3.12: The SAFAD results, screenshots from the SAFAD website [32]

4 Discussion

4.1 Key takeaways from the user feedback survey and the analysis of available calculators

Based on the user feedback survey, tool A - the *Free food carbon footprint calculator* scored the highest for user-friendliness, while tool C - “*How does what you eat affect the planet?*” scored the highest for usefulness (3.3). Additionally, users voted tool C as their most preferred tool overall. This feedback guided the focus toward understanding what users found well-executed in tools A and C, as well as their preferences for ease of data input and result visualization. Users expressed a preference for the future tool to have a less cluttered interface, citing concerns with Tool C, where the presence of too many elements on a single screen left them feeling overwhelmed by excessive information. They appreciated the interactive features and emphasized the importance of maintaining an aesthetically pleasing design. At the same time, they desired a tool that is straightforward to use, with a relatively simple data input process. Finally, users expressed a strong interest in seeing results that go beyond climate change impacts to include other environmental impacts, all within a tool that remains easy to navigate and understand. The analysis of available tools, including tools A, B, C, and the SAFAD website, revealed several desirable functionalities and usability features for the future tool. Similar to tools A and B, the future tool should include a designated section for users to input recipe ingredients along with their corresponding weights or portions. Additionally, like Tool C and the SAFAD website, the results should present multiple environmental impact categories preferably in graphic representations.

Building on this user feedback and the analysis of existing calculators presented in 3.2 and 3.4, the proposed standards for the tool’s functionality and usability aim to reflect these preferences. They incorporate key user desires while leveraging the strengths observed in the evaluated tools, ensuring a comprehensive and user-centered approach to design.

4.2 Standards for the prototype’s functionality

The final tool must include a wide range of environmental impacts to provide users with a holistic view of the impact of the chosen recipe beyond GHG emissions. The tool should measure other environmental impacts such as acidification and land use that align with the work done by Barco Alzate. To make the tool dynamic and interactive, features that could provide real-time feedback should be included, as those were appreciated by users. For example, sliders for adjusting certain ingredients should allow users to immediately see the impact of their recipe changes. Users should also be able to explore detailed breakdowns of environmental impacts by food group, ingredient, or preferably lifecycle stage, offering an in-depth understanding of the data.

Customization will be a core element of the tool’s functionality, allowing users to input their own recipes and tailor ingredients to their specific needs. To support sustainability, the tool should also suggest alternative ingredients that lower environmental impact while preserving

nutritional value. Furthermore, it should present contextualized results by translating raw data into relatable equivalents, such as real-life activities, to make the impact more relatable and easy to understand.

The key functional requirements and recommended feature lists were derived based on an analysis of user feedback and prior work. However, their inclusion in the future tool itself can depend on the technical complexity of coding and implementation, thus a few modifications may occur in the future work.

Key functional requirements for the future environmental impact calculator

- The calculator's results should include a wide range of environmental impacts beyond GHG emissions, providing a comprehensive and holistic assessment of environmental impacts.
- Include ingredient breakdowns in environmental impact result graphs to offer an in-depth understanding of the data.
- Present results in relatable terms by converting raw data into real-world equivalents, making the results easier to understand and more meaningful for users.
- Users should be able to select ingredients in various units or portion sizes appropriate for each ingredient type.

Recommended functionality features for the future environmental impact calculator

- Including pre-made meal options with editable ingredients would make the tool easier to use, allowing users to modify existing recipes rather than starting from scratch.
- A feature suggesting more environmentally sustainable ingredient options, or limiting meal frequency to encourage sustainability while maintaining nutritional value, would be beneficial.
- Real-time visualization of impact changes as users adjust ingredient amounts will help them immediately understand the environmental consequences of their choices.
- Providing portion recommendations for ingredients, as seen in tool A, during the selection process could help users make more informed choices and ensure accurate input.
- Inclusion of an environmental impact rating (Low to High) would help users quickly grasp a recipe's impact in comparison to others.

4.3 Standards for the prototype's usability

The UI of the tool should be intuitive and well-organized, with minimal clutter to avoid overwhelming users. Features such as a search bar, drag-and-drop functionality, and food icons could make navigation simple and efficient. To improve the user experience further, the tool should offer flexible input methods, allowing users to enter data using both portion sizes and weight measurements. As portion size suggestions from tool A were praised by users, this feature could also be useful to include in the future tool.

The tool should prioritize effective visual representation of results, using graphs and charts to make complex data easy to understand. Options to toggle between aggregate and detailed views should be provided, allowing users to explore the level of detail that suits their needs. Relatable comparisons, such as emissions equivalent to driving distance, should be incorporated to make the impact data more tangible. Accessibility will be another key focus, with the tool designed to be responsive and functional across devices, including mobile phones. Integration with grocery shopping apps could further enhance usability by allowing users to analyze the environmental impact of their purchases in real-time. These can be discussed further next semester in the final phase of the Master's thesis project.

Similarly, the key requirements and recommended feature lists for the future tool's usability aspect were derived.

Key usability requirements for the future environmental impact calculator

- A clean and simple design with an easy-to-use interface should be prioritized to avoid overwhelming users.
- Incorporate graphic icons for different ingredient groups for the ingredient selection process.
- Incorporate interactive icons for environmental impact results, allowing users to click and explore specific categories for their explanations.
- For the ingredient selection process, the ingredients should be neatly categorized for easy browsing, with a search bar enabling users to quickly find specific items.
- Use a clean, aesthetically pleasing color palette to maintain a visually appealing, uncluttered design.

Recommended usability features for the future environmental impact calculator

- Adding an option to save or track recipes over time would help users build habits and easily reference past results.
- The tool should be responsive and accessible across various devices, including mobile phones, ensuring it is usable in multiple contexts.
- Integration with grocery shopping apps would allow users to assess the environmental impact of their purchases in real-time, enhancing convenience.

These standards provide a framework for the design and development of the final prototype. By addressing user preferences and incorporating the strengths of existing tools, the final design will ensure a balance between functionality, usability, and user engagement, creating a tool that is both effective and accessible to users with varying levels of sustainability knowledge. These key features and standards will be implemented in the next phase of the project to design a user-centered environmental impact calculator.

5 Conclusion and future work

The user feedback survey emphasized the importance of a tool that balances detailed environmental impact insights with simplicity and accessibility. Users valued features such as dynamic feedback, sustainability suggestions, visual clarity, and regional customization. These preferences, combined with the analysis of existing tools, have informed a set of standards for the final prototype's functionality and usability. These standards prioritize a clean, intuitive user interface, flexibility in data input, various environmental metrics, and features to engage users through easy-to-understand graphs and relatable comparisons.

Looking forward, the next phase of this project, planned for the upcoming semester as part of the master's thesis, will focus on the design and development of the tool. This design phase will implement the defined standards and key features to create a functional prototype that meets user expectations along with preferences, and project goals. The design process will aim to translate the feedback and insights gathered into a visually appealing, interactive, and impactful tool that empowers users to make informed and sustainable dietary choices.

By building on the established standards and leveraging user insights, the next semester will represent a critical step toward creating a tool that calculates environmental impacts and encourages users to take actionable steps toward a more sustainable future. The resulting prototype will lay the foundation for further refinement and testing, ensuring its readiness for broader implementation.

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A User feedback survey



Environmental Impact Calculator Design

Thank you for taking the time to participate in this survey!

This form is part of Jeongmin's thesis project, focused on designing an effective User Interface for the Food Recipe Environmental Impact calculator with an emphasis on evaluating data contextualization.

This project is part of the EU-funded RAINFOREST project (link: [RAINFOREST](#))

- 🕒 The survey should take about **15-20 minutes to complete**
- 💻 It is **preferable to complete it on a laptop or desktop** as you will need to open and interact with some external websites during the process.

Your feedback will be essential in shaping the final prototype, helping us identify useful features, areas for improvement, and how to create a user-friendly, impactful tool for understanding the environmental impact of food choices.

We greatly appreciate your time and input. Thank you for your participation and support in this project!

Data usage disclaimer: Your responses will be used exclusively for the thesis project and will remain confidential.

[Logg på Google](#) for å lagre fremdriften din. [Finn ut mer](#)

*** indikerer at spørsmålet er obligatorisk**



What is your current background or sector of work/study? *

- ☐ Student
- ☐ University/Academic
- ☐ Government
- ☐ Private sector/Industry
- ☐ Non-Governmental Organization (NGO)
- ☐ Others
- ☐ Prefer not to answer

What is your age cohort? *

- ☐ 10-20
- ☐ 21-30
- ☐ 31-40
- ☐ 41-50
- ☐ 51-60
- ☐ 61-70
- ☐ 71 and above
- ☐ Prefer not to answer



What is your gender? *

- ☐ Male
- ☐ Female
- ☐ Other
- ☐ Prefer not to answer

What is your nationality?

Svaret ditt _____

What is your place of residence?

Svaret ditt _____

Have you used any environmental impact calculators before? *

- ☐ Yes
- ☐ No
- ☐ Prefer not to answer



How familiar are you with various environmental impacts, such as greenhouse gas emissions, land use, eutrophication, etc.?

*

- ☐ Very familiar
- ☐ Somewhat familiar
- ☐ Not familiar at all
- ☐ Prefer not to answer

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* indikerer at spørsmålet er obligatorisk

You are now asked to explore three websites that calculate environmental impacts. Please take your time to explore each one and try calculating the impact of different recipes or diet styles that interest you. Here is the **Tool A**:

[Tool A - Free food carbon footprint calculator](#)

What did you like the most about the tool A?

Svaret ditt

What did you dislike or find challenging about the tool A?

Svaret ditt



How user-friendly did you find the interface of tool A? *

(User-friendly means *easy to use or understand* & Interface is a way for users to interact with the website)

Very difficult to use 1 2 3 4 5 6 7 Very easy to use

☐ ☐ ☐ ☐ ☐ ☐ ☐

Did the tool A provide clear and useful results on the environmental impact of food? *

No, not clear 1 2 3 4 5 6 7 Yes, very clear

☐ ☐ ☐ ☐ ☐ ☐ ☐

What did you like about how the results from the calculation were shown at the end?

Svaret ditt

What additional feedback do you have regarding tool A?

Svaret ditt

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You are now asked to explore three websites that calculate environmental impacts. Please take your time to explore each one and try calculating the impact of different recipes or diet styles that interest you. Here is the **Tool B**:

[Tool B - Carbon Calculator](#)

*The security certificate for the site has expired, and while the website is still accessible, it may not be secure. **You can proceed at your own risk**, but please be aware that the site may not protect your information.*

What did you like the most about the tool B?

Svaret ditt

What did you dislike or find challenging about the tool B?

Svaret ditt



How user-friendly did you find the interface of tool B? *

(User-friendly means *easy to use or understand* & Interface is a way for users to *interact with the website*)

Very difficult to use 1 2 3 4 5 6 7 Very easy to use

☐ ☐ ☐ ☐ ☐ ☐ ☐

Did the tool B provide clear and useful results on the environmental impact of food? *

No, not clear 1 2 3 4 5 6 7 Yes, very clear

☐ ☐ ☐ ☐ ☐ ☐ ☐

What did you like about how the results from the calculation were shown at the end?

Svaret ditt

What additional feedback do you have regarding tool B?

Svaret ditt

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You are now asked to explore three websites that calculate environmental impacts. Please take your time to explore each one and try calculating the impact of different recipes or diet styles that interest you. Here is the **Tool C**:

[Tool C - Planet-based Diets](#)

What did you like the most about the tool C?

Svaret ditt

What did you dislike or find challenging about the tool C?

Svaret ditt



How user-friendly did you find the interface of tool C? *

(User-friendly means *easy to use or understand* & Interface is a way for users to interact with the website)

Very difficult to use 1 2 3 4 5 6 7 Very easy to use

☐ ☐ ☐ ☐ ☐ ☐ ☐

Did the tool C provide clear and useful results on the environmental impact of food? *

No, not clear 1 2 3 4 5 6 7 Yes, very clear

☐ ☐ ☐ ☐ ☐ ☐ ☐

What did you like about how the results from the calculation were shown at the end?

Svaret ditt

What additional feedback do you have regarding tool C?

Svaret ditt

Tilbake

Neste

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Comparison and Preferences

[Tool A - Free food carbon footprint calculator](#)

[Tool B - Carbon Calculator](#)

[Tool C - Planet-based Diets](#)

Which tool do you prefer the most? *

- ☐ Tool A
- ☐ Tool B
- ☐ Tool C
- ☐ Prefer not to answer

Why do you prefer the chosen tool above the most?

Svaret ditt _____



Was there a feature in any tool that you found particularly helpful? If so, which one and why?

Svaret ditt

What feature(s) were missing from any of the tools that you would like to see in a new environmental impact calculator?

Svaret ditt

Tilbake

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Interface and Features

Would you prefer the results to be displayed as: *

- ☐ Visual charts/graphs (e.g., bar graphs, pie charts)
- ☐ Simple numbers/percentages
- ☐ Comparison to real-world activities (e.g., "this is equivalent to driving X km")
- ☐ Prefer not to answer

How easy should it be to input data (e.g., selecting ingredients and their weights) *

- ☐ Very simple (minimal input)
- ☐ Moderately simple (some flexibility)
- ☐ Advanced (customization, full manual input)
- ☐ Prefer not to answer



Would you like the tool to offer suggestions for more sustainable ingredient alternatives in your recipe? *

- ☐ Yes, definitely
- ☐ It would be nice to have
- ☐ No, not necessary
- ☐ Prefer not to answer

What user interfaces or result presentations do you feel are missing and would like to see in the future?

Svaret ditt

Tilbake

Neste

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Environmental Impact Calculator Design

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* indikerer at spørsmålet er obligatorisk

Open-Ended

If a calculator with your desired features were available, how often would you use ^{*} it?

- ☐ Daily
- ☐ Weekly
- ☐ Monthly
- ☐ Occasionally
- ☐ Rarely
- ☐ Never
- ☐ Prefer not to answer

Do you have any other comments or suggestions about the calculators or the survey?

Svaret ditt



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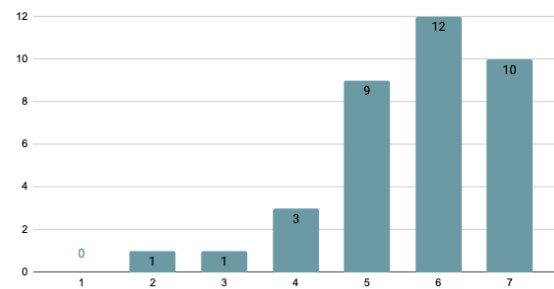
Google Skjemaer



B Complete results of tool A and tool B

How user-friendly did you find the interface of tool A?

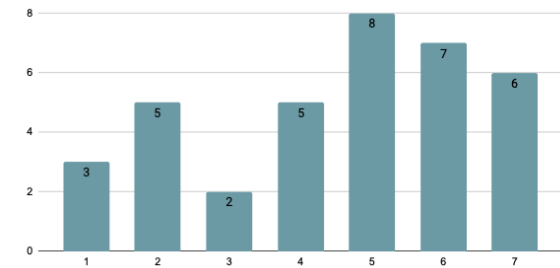
36 responses



(a) Tool A's complete user-friendliness results

Did the tool A provide clear and useful results on the environmental impact of food?

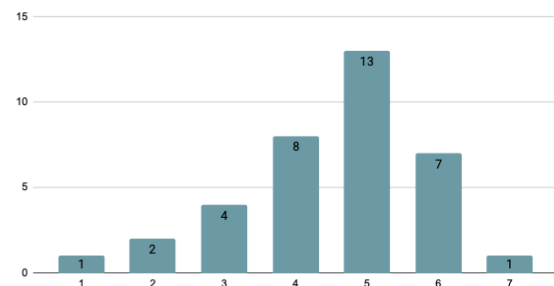
36 responses



(b) Tool A's full usefulness results

How user-friendly did you find the interface of tool B?

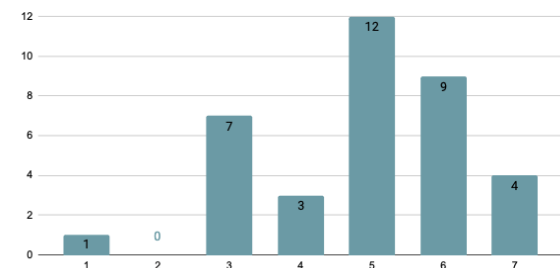
36 responses



(c) Tool B's full user-friendliness results

Did the tool B provide clear and useful results on the environmental impact of food?

36 responses



(d) Tool B's full usefulness results

Figure B.1: Complete results on user-friendliness and usefulness of tool A and tool B, screenshots of the Google survey results

C User experience and feedback results

C.1 Tool A

What did you like the most about the tool A?

Easy to use and to understand
that it gives weight hints for typical servings - as it can be hard to asses on your ownhow much does a portion weight
I liked how it said the typical serving size
That they had suggestions for typical serving sizes of the different food types!
Easy to use
Drop-down search
they have a long list of ingredients, and suggest a common serving mass
It has nice colors and is easy to use. I like that I can compare the ingredients and also see how it compares to transport in a year and other foods.
Quite intuitive. It was easy to find the ingredients in the list. Also liked that you could choose different units.
simplicity
Simple. Nice to have typical serving suggestion.
Easy and straightforward
Very intuitive with simple and efficient user interference. I also liked the comparison to other activities
Easy to search through food types and interactive
Nice UI, lots of options ingredient options to chose from. Can chose different measurement units (not forced to calculate in cups, ounces or pounds).
Ease of use and options
The "standard serving" weight approximation
Not sure
Easy to investigate different ingredients
the amount of ingredients and the description of how much is a serving as well as the ability to add multiple ingredients in a calculation
Nice choice of foods and suggested serving sizes.
Simplicity, suggestions about typical sizes
It is an amazing calculator. I wish if the food products have such labes written on the packaging, just like the engergy labels on the household appliances. Since, even though this is a very useful tool, one might not often use them for every individual buy.
Really liked the suggestion of a typical serving with corresponding grams! Helped a lot!

Amount of choice
Easy to use and has most foods. I like "equivalent of" option, helps put things in perspective. I also like that it helps you estimate how many grams by giving you small tips.
N/A
How simple it seems
It didn't work, but it looked like a nice layout
Easy to use
When I go to fill the amount in grams, it gives me a quick comparison with other measurements. Like, for entering grams for a spice, it would tell me that 1 teaspoon is 3 grams, which helps.
It is neat and easy to navigate. Resources menu is interesting
Simple to use. Straightforward label. Easy to interpret
Easy to search for ingredient. Fast answer

What did you dislike or find challenging about the tool A?

didn't find anything about different types of meat?
Nothing really! Really user friendly and good options of food types! Could be more detailed than just GHG-emissions
create combination with other ingredients which are used to prepare meal. also the input unit is only gms which is not user friendly for liquids
Focus only on emissions, while land and water use related to agriculture are biggest drivers of biodiversity loss
takes time to enter all the ingredients from your meal, if you had already prepared meal it would be easier
At first I did not see you had to enter a number of servings (even though in big at the top...), so then when the results arrived and it was "gCO ₂ eq per serving", I was wondering "what is a serving?". It was not clear the comparison of ingredients: it is written gCO ₂ eq, but it was not straightforward that it was a contribution analysis based on the quantities I entered (a per unit comparison would also be interesting to have - without entering 1 for all ingredients).
search option in the dropdown was not obvious in the beginning. After realizing, adding a recipe went much faster
Nothing
Only CO ₂ emissions were included

It would have been nice to compare my result to the average consumption to have some hooks I could relate too (but I guess you could just switch around the ingredients)
Not sure how to see the system boundaries or how the results were generated. Black box?
Would be nice with premade dishes (or blueprints), so don't have to enter everything manually. Don't really understand the result (what does xxx gCO ₂ e per serving mean or correspond to).
Categories can make it easier to use
Food category selection
I think the comparison is flawed by design -- my muesli portion is compared to pizza slices, and my first thought is "wtf, one muesli serving is not equivalent to three pizza slices". Which means WHAT is being compared is not clear enough. The emissions value gCO ₂ e is not clear to an average user. The tool tilts them towards eating unhealthy food (say, baguettes) based on the urge to get to the lowest emissions rating possible. It also creates a feeling of guilt over eating healthy food by comparing the serving to travel by car or train. So basically if I just drive by car for 150 km a week, I can skip eating? :) Or maybe we should all eat baguettes with A rating and skip fruits and vegetables altogether? Finally, the tool completely ignores the origin of the food. What if I grow my salad in my home garden and import baguettes from France on weekly basis?
I am worried it does not take the transport/season of ingredient into consideration
that only produces results in terms of CO ₂ eq. The process of adding ingredients is slow, since it's a very long list. can be complicated to navigate
COuld have been nice to feed in menus with 3 course meals
If it didn't have the ingredients I want to add, I have to poke around to find something similar. Also that it only shows the carbon footprint but not other environmental impacts.
As I said above it is a tedious task to search for each and every food, scroll down and look for the energy labels. The good packaging should have carbon food print labels. However these kinds of tools can be improved with AI, for example as we type it can show the food type and its foot print.
Several spelling mistakes in the resources page.
No explanation of what gCO ₂ e per serving means
Values based on estimates, but it does not say anything more about where these estimates come from, production method of the food, and which country the data is from. Maybe it does somewhere on the website, but its not immediately clear.
The tool didn't let me add ingredients, so I could not try tool A
Doesn't seem to work. And I assume it would take a while to put in an entire recipe
Maybe it could be a little text explaining what to do? And also the colours were not super inviting. I'm not sure why but the combination didn't appeal to me in the context of food.
Would like an explanation of why some ingredients have high carbon footprint and some have not
it is a bit difficult to make complex recipes, like those from original mexican food, mainly due to lack of ingredients
Nothing in particular

Did not have what I was looking for

What did you like about how the results from the calculation were shown at the end?

Very clear to read
i like that you can see equivalent products and compare if you added more than one
That it said "this is a low footprint" and similar conclusions
I really like that they compared it to other food types or travelling distances.
the co2 emission level like high or low
Aggregate value and breakdown by ingredients is useful
it says if it is a low or high carbon footprint
It was short and concise, with nice describing illustrations
Interesting to have the equivalent to other food. Interesting to also have the equivalent to other activities (traveling).
comparison on a scale
Gave a carbon rating, equivalents and compared the ingredients
It showed whether the carbon footprint was high or low
Visually pleasing. Easy to understand what was relevant. I also liked that you could look into which ingredient had the most impact.
Yes, but maybe more disaggregated results can be more informative
Nice big number, but not that much explanation of what it means.
The scale
The "high-low impact" scale
I like that it can add together a list of different ingredients and that you can compare them
it was easy to understand
Clearly see beef and such have high impact compared to dairy and vegetarian choices
The possibility to compare ingredients.
The results shown are impressive, however only the necessary mode of transportations are shown as comparison. Not the bombs we drop at each other in the war or conflicts, or to save democracy. Maybe we can include the equivalent of how much war ammunitions are produced and used all over the world to defend each other, something which is missing from every environmental reports or analysis.
Very clear with showing only one number.

The A to E scale and the "equivalent to" tab
N/A
Doesnt work
I could run anything because the site was down
Clear what the footprint was
I like the ingredient comparison at the end, which tells me of all the items which is the highest contributor and how it compares against others.
I didn't like it much. I would like to know a lot more than just a number, like what contributes the most to the impact or what the how the number for each ingredient is calculated/comes from.
Traffic light system. Easy to interpret.
Color and words (high/low), not only numbers

What additional feedback do you have regarding tool A?

Just CO2eq, other impacts would definitely be needed (like water consumption).
highlighting the most and least impactful ingredient could be beneficial
Could have presented all the results at once, instead of clicking on equivalents and comparison of ingredients.
More impact categories would be better
Would be nice with some comparison what the emissions correspond to.
No uncertainty
Guilt-inducing tool that simplifies way too many factors.
A bit hard to find ingredients. Tried just writing apple and it did not find it, I needed to scroll past other versions and click "Apples" in the list. Also stores your previous and non existing attempts in the list, so tried clicking my old "Apple" and "apple" which did not exist.
it would be nice to see more information on what the result means, how it compares to other diets or activities, what the consumer can do to lower the impact of their diet, and how it compares to other impact categories. Is it good in terms of CO2 but bad for biodiversity? bad for use of land? water?
Have option of providing impact of recipes from links

It is a great effort and I like the tool very much. Specially it shows how meat eating is affecting the enviornment. One should have these kind of tool in every mobile phones. But please include the war ammunition manufacture and usage's CO2 footprint in comparision in these kind of tools.
I tried inserting both a cup of coffee, which got a high carbon footprint. I retried with a bread slice and jam, which got a higher number but said that the footprint was lower. That could be confusing.
Wish it gave a more detailed breakdown on how each ingredient contributes. Maybe it does and I just missed it, idk
Doesnt work
nothing
I would guess the footprint depends on where you are form, what the country can provide itself and what has to be imported, but does not say anything about that
Would be cool to add ingredients and sum all up. Sand then sum to compare with two or more meal options

C.2 Tool B

What did you like the most about the tool B?

It was very detailed and it gave me a comparison on the emissions
nice that there is step by step instruction on how to fill in the gaps
That it gave an alternative example of how big the carbon footprint was
Cool to be able to chose region of origin, and more detailed in that sense.
selection categories and measuring units are better than tool A + origins of the foods
That I could choose the region of origin
they have the regions where the ingredients come from
It includes region of origin
Having the region of origin.
Comparison to car driving makes results tangible
Nice to have region of origin
Very regionalized
The attempt to letting you add details such as country of origin.
Easy to use and search through selections. It helps so much to be able to select regions too
Nice that it shows how much emission corresponds to in real life use (driving xx miles in average car). Gives meaning to what the emissions mean.
The detail it provides for options
Region of origin question
One screen UI!
Takes region of origin into consideration
that the ingredients are grouped by categories
Easy to see the available ingredients
It gives the carbon footprint of total meal.
Very easy and intuitive to start the calculation. Nice to have the comparison with diving a fuel car.
That it has region of origin
easy to use
How it seems to withstand large recipes without being confusing.
I liked the colour combination, and it was also very intuitive
the page would not open....

It is simple
Sorting by categories (beverages etc.) makes it easier
Did not manage to access it. ("The security certificate for the site has expired")

What did you dislike or find challenging about the tool B?

It did not have Feta cheese and it took a bit more time to fill out
providing weight of products
The design didnt feel very clean
Not as easy to use as the other one, and looked "cheaper" and not as fancy. No suggestion for normal serving sizes :/ Also missed more comparisons of different travel methods and different food types.
no calculate but at the beginning
While the ingredient categorisation is helpful, it requires more clicks than a simple search
don't have a lot of ingredients could not find all ingredients from the same meal I used in tool A
The page is not divided well, it is all the same color so difficult to see what is what. It was not so clear how to calculate compared to tool A. It can also be confusing that the total per serving and the total overall has different units. Very few illustrations made it feel more "scientific", like I am doing research and not just checking my footprint as an individual.
Suddenly much more text to read (less attractive if one wants to do a quick calculation) and the user interface is not very nice (a table for the recipe listing which does not look very good). Less choices of units: no ml (which is not convenient if you are using cup to bake or for liquids such as quantity of oil or milk)
The option to search was not made obvious. This is much faster than using the dropdown
Bad UI, too many things to click on.
A bit difficult to find the ingredients. For example "chicken" was not easy to find
I didn't like that things seem to pop up as you filled in new information. Boxes moved around on the page, and it didn't give me anything to compare to either. It was also weird that the dropdown menu went upwards, not downwards.
Not possible to see the disaggregated results and system boundaries
UI is way worse. Difficult to find ingredients and couldn't find all options.
Too colorful
Amount based on weight is always difficult
All the dropdowns, many, many categories. The environmental impact info is meh. What if I don't own a car? Why is the tool so UK-centric?

Plain numbers of Co2 emissions does not tell me much.
add the amounts is challenging because you need to change the units and add the amounts without guidance of how are the servings in average
Very limited list of available ingredients. Several things, incl. regions of origin, seem very UK centric.
Why always compare with transportation? Which is an essential/need. Why not compare with non-neccessary carbon footprints like war ammunitions , building statues, making luxury electric cars?
Not as easy to know if the footprint is good or bad.
That it asks for amount of servings. Was confusing at first but then it just seems that the total is divided by the number of servings, which is pretty straightforward maths.
limited selection of foods
Finding the food i want.
It looked like it was not yet finished, but that is perhaps an in process thing.
Can't open the site, SSL Certificate expired. Error message "Your connection isn't private Attackers might be trying to steal your information from assets.plateupfortheplanet.org (for example, passwords, messages, or credit cards). Learn more about this warning net::ERR_CERT_DATE_INVALID"
A bit difficult to figure out how to start
Nothing in particular
Did not manage to access it.

What did you like about how the results from the calculation were shown at the end?

Liked the comparison and the per serving section
the comparison to car driven distance
As said previous the driving car comparisent
Compared to driving with a petrol car. Also nice to see the emissions per kg of product in parenteces.
less clear idea on the level of carbon emission than tool A
That the result is compared to an everyday activity
you have a sentence comparing your emissions to a driving distance
Easy to read by having per serving and the total
Comparison to car driving
Nothing in particular

It gave an example of what the impact means
Clear result, and good thing to compare it to a car, for example (even though that's not a very good comparison, since that covers two widely different needs for a person)
I like that we can see the equivalent of emissions in simple terms or practical terms
Easier to understand compared to calculator A
Comparison with car emissions
I like that it skips to the essence and doesn't give you all the needless and dumb comparisons to number of pizza slices etc.
Easy to change amount in the list. See how this could be used by restaurants etc to add info to menus
it was clear
Need more non-essential comparisons. Otherwise people think that just using paper straws and electric cars will solve the environmental issues.
Could be more appealing maybe, with more colours or visualizations.
I guess that per serving can be a useful value to have when making large meals.
Good. Some more comparisons would be nice. How much a car emits is just another thing that is a bit hard to understand.
I liked that it compared my result of eating tomatoes with the amount of miles in a car you have to drive, that was a fun aspect.
I liked it gave an example of the impact in driving miles with a petrol car.
Reference to miles driven, but i think traffic light works better.
Did not manage to access it

What additional feedback do you have regarding tool B?

it states that it considers countries of origin, but you can just choose region, or UK, and back to site bottom is not working... you have to restart the website to start accounting again
Put a km in the result as well, not everyone uses miles :)
Liked tool A better
need to be able to add other cooking ingredients and cooking time / method for real life emission
Again, focus only on carbon can be misleading
add already complete meal
The comparison is in miles for UK (not very easy to understand when one is not used to that unit). Also same as tool A, just GHGs.
I liked that the number of servings was not pre-filled. It's something easily overlooked and impacts the result hugely.
It would be better if the ingredients can be easily selected
I struggled a little bit to use it correctly since I first tried to write some food that apparently didn't exist in the database. It returned good results, but for someone who doesn't know much about the general CO2-equivalents of food per serving, it is hard to make that number matter. I also missed the comparison of the different ingredients as visually pleasing as the last tool.
it can also provide a detailed emission per ingredient if requested
Could be more straightforward selection of items
the use of the tool is a mix between user friendly and overwhelming
Would have been simpler to feed recepies and some of the off the shelf products to make the point of environmental impact and sourcing from various locations
Disturbing that the image at the top didn't load and clicking on it lead to a broken site. Unmaintained?
The tool is showing that petrol car is the evil to the environment. And not the luxury electric cars, more babies, bombing each other etc.
No additional comments :)

That they fix their logo
Give a search option, or what would be even better: an AI feature where you can type in the name of a dish, and it assumes a certain set of ingredients.
Maybe make automatic unit choice depending on the category (eg. beverages should be in litres not kgs)
Did not manage to access it

C.3 Tool C

What did you like the most about the tool C?

It was very detailed on both endpoint indicators and inputs
It's more complex but still easy to play with - I like that it suggests the average diet for each country and you can easily modify scenarios without putting in specific products and their weights. Also it's really nice that while making your options it sums it up to kcal/day and marks if it's optimal for health
Seeing the typical norwegian diet
Several different impacts! And showing compared to planetary boundaries and including calories and health. Easier and quicker to select the total portions per week.
full diet plan is covered, all type of environment impacts are shown
Sliders
very different, gives you a better picture of your whole diet as well as suggesting you to eat enough calories
Includes information about typical diet for my country and interactive graphics. Also nice that it includes information about health.
The different impacts and the planetary boundaries. Interesting to have the human health aspect too. I liked that you could choose a country, see the typical diet, and compare your diet with it.
Direct feedback from using the sliders to seeing the results without hitting a button and being able to use the average citizen as a benchmark to start with.
A lot of detail and interesting information. Could see typical diet.
It gives results for more impact categories than climate change
I liked to adjust the portions and see how that impacted human health and the planetary boundaries. I also liked how it made suggestions for you to what you can do better to make the diet work.
The feel of the website
Really nice user interface. Like how it has presets for countries. Easy to see results.
It can give you a whereabout about your country
Portions per week description (maybe more relevant than individual food items), "optimal range" indicator, different impacts, "typical diet" for country option.
The slider system and the live-updated impact ranges. The clearly explained portions. I would just like to sit and click through the different countries to see what their diet looks like, how it compares to the world diet etc. Very nerdy tool. I feel this is more useful than other tools. I can see how, say, eating less meat can help the environment, how I can balance my diet to help both myself and the planet.
Fun interface, you see different categories of impacts
the amount of information on each step
Categorization of land use
Gives very complete picture, on a whole diet with regard to several environmental impacts. Also picking

a typical diet for my country was a nice feature.
Awe! this is the best tool, a user will go to total plant based diet. I love it.
Initially liked that you could just add portions instead of thinking about the weight.
Easy overview
The impacts tab is beautiful. Also liked the optimal range
easy to use
Very easy to use and get a certain understanding of.
I liked the colour combination, and that I could move the sliders
Easy to use and country based
I like that I can choose typical diet from different countries and see their impact.
I loved it. It allows to create/calculate a diet in the context of personal health, which is what matters the most. The impact is also measured in the context of how the diet impacts the food system in the contry you live.
Planetary boundaries!
working with portions (not kg) and easy categories, quick to use

What did you dislike or find challenging about the tool C?

It was hard to use and to understand. The only two options i got was "you greatly exceed the plantes boundaries" and "you are eating sustainable" no in between
the impact value is not visible right away - but it can also be an advantage for an average user
Accepting that norwegian diet sucks
Nothing really. Loved it
not really but might be a bit complicated for regular people to understand (who are not familiar will all type of climate impacts)
The page appears very cluttered
too much to think about, a little overwhelming when you first enter the site
Less intuitive to know what a serving is than comparing actual weight of food like in the previous tools. It feels more "noisy", takes more time to understand the tool.
The human health aspect is not adapted to the morphology of a person.
Initially it looked like eating nothing is promoted as the ideal solution. The disclaimer that you still should fulfill your caloric needs could be even bigger and also on the results side!
Not intuitive
The ingredients are not possible to be specified. No regionalization of where things come from
You had to set aside more time to read about the definitions of a portion, and the food category. I like

better to differentiate with mass, rather than portions.
Not easy to see how to select specific food types
The consumption is listed in "portions". Don't really know how to measure in portions, or what it corresponds to. What is "30 portions per week of fats & oils"?
Not easy to calculate your portions by yourself
Hm, can't think of anything.
Carrots is in vegetables category, but potatoes is not. A bit confusing with overlapping categories. Portions also a bit difficult to calculate and requires reading up on what counts as a portion for each category.
nothing
A lot of effort to insert a realistic diet, I would have to look up a lot and so I didn't bother.
It still shows fish, meat, dairy can be consumed in large portions, which is the main source of carbon emission after the war ammunitions. Not the basic necessary transportations like cars.
Hard to know how many portions of sugar I eat a week. Did not understand how the portions worked and ended up with too few calories.
What is a serving exactly?
Using number of servings can be a bit tricky to wrap your head around. Also a whole lot of categories you need to think about
It doesn't really help getting a feel for how certain ingredients or food items impact. A mix between this and something that breaks it down into specific sample dinners typically eaten would be great.
The sliders where not in the same number range, maybe change it so that the same placement equals the same value.
challenging to get within the recommended range
A bit difficult to use. Too many food categories. May be could be simplified. Also calories per day should be possible to adjust based on activity level and gender?
Showing the score while choosing was distracting. And the recommendations were reducing dairy products even when I only increased alcohol.

What did you like about how the results from the calculation were shown at the end?

It showed what was specifically affecting my emissions and also in which midpoint category i was emitting the most. Makes it easier to adjust what i am eating and understand the impacts.
I liked that there are set some boundries and it suggests that you should limit consumption of given products
Showed how much is consumed contrary to how much one should consume
Greatly presented the results compared to boundaries. Great with the recommended range and showing calories as well. Really cool that you could press the different impacts and see the percentage each food type contributed.
overall climate impacts and somewhat real life comparison to make the impact easily understandable, the circle graphs
overview of multiple impact categories
you have graph and thresholds on different planetary boundaries as well as comparing from the average in the country you live in
Its clearly shown in a nice graphic and it has information about health impacts and information about diet shift.
The variety of impacts + planetary boundaries. The results are also displayed in a very interactice way with changing some of the inputs which changes the results in real time. Very cool to see what you can change and how much impact it has. I like that you can click on the impacts and get to know more about the contribution from the different items to each impact.
More impacts than just CO2. Benchmarks about bearable impact.
Good visuals and interesting details.
I like that it shows what is the limit and whether the user is exceeding it. It shows what is needed to be targeted
I liked how they gave a lot of possibilities to see the actual impact of what you eat. How can you increase the calories while still keep within the planetary boundaries, for example. That actually provided some useful insight to how I can improve my diet to work for both me and the planet.
National level results
Easy to understand, and nice how it breaks everything down into what contributes most to each category
It gives more detailed impact indicators
Same as above.
See the answer to what I liked most.
Nice graphs. Easy to see what was considered good with "threshold" values to try to stay below.
they are clearly explained and it's possible to compare with other categories and includes references and

more resources
Celebrations when meeting environmental goals
A lot of background information provided, if I want to read them. Suggestion about what I can change to effect what aspect.
They look good
Liked that it gave nformation on different impacts, such as biodiversity and water use.
The impacts split into different categories and also warnings when you exceed recommended limits.
Very intuitive and clear.
Some part of the categories were unrecognisable, as it wasn't possible to scroll all the way down. but I liked that it was shown as a thumbs up indicating that you are within boundaries.
Results shown in an easy way to read and understand
I like the classification for impact.
Results are given in terms of Human Health and the impact to the local food system.
Reference to planetary boundaries is super interesting.
Nicely divided in a few easy categories, not to much but still more than CO2

What additional feedback do you have regarding tool C?

Could add more nuanced feedback when exceeding the limits.
Overwhelming making your own diet, not sure how it can be reduced tho...
Lovely
nothing much
While useful to get an overview of a full diet, it is very difficult to break down insights to individual meals
The tool is mostly focused on diet and not for a specific recipe. Being able to choose between evaluating a diet or a recipe could be nice.
A design that would not require scrolling on the input-side would make it even easier to use. Having to scroll up and down is bad for keeping an overview.
Adding ingredients in details would be better

More info on average countries of origin would be nice. Show results at bottom of page.
Would be nice to feed pre-made food options and also know the impact of packaging
May be show the meat/dairy/fish in red color?
Looked very nice at first but found it very hard to relate to my actual consumption.
They really hate my diet
The only thing this tool can give me tips on is whether or not I should eat less meat etc. It doesn't say anything about what impact certain ingredients for instance has. But maybe that is of no big importance to an average consumer
This was my favorite of the three. I wasn't exactly sure how to add type of servings and it looked like I never was within the calorie boundaries.
The menus are also interesting.

D Comparison and preferences results

D.1 Reasoning for preferred tool selection

Why do you prefer the chosen tool above the most? - Tool A

Easy to understand and use
Seems the easiest to use and the cleanest
Search field
Simple for someone who is not very interested
It stops at the time of household purchase. Giving early warning!! But if it is combined with the Tool C it will give amazing results.
Was easy to understand and gave good support when filling in information.
I liked A the best if you are calculating for a single meal. For calculating the impact of my entire diet tool C would of course be best fit.

Why do you prefer the chosen tool above the most? - Tool B

Easy and regional-level result

Why do you prefer the chosen tool above the most? - Tool C

its easier to use and has the most friendly interface
Several impact categories. Most detailed, while still being user-friendly!
covers the most of the climate impacts + kalori intake suggestions
although it is different from the other 2, it is the one that can really paint well a good picture of you diet and its impact
It was not the most intuitive to use, but I think it is important to include information about health.
Several impacts, planetary boundaries, interactive.
Most information and best input-output connection.
It has more impact categories than only climate change
Difficult to choose between A and C. A had a very nice user interference and you simply had to fill in some boxes, but C gave much more insight, though I'd prefer to rely on mas rather than portions.
User interface, ease of use, presentation of results. Nice colors and graphs.
more impact categories, sliding scale, long-term dietary Outlook.

Lots of data, clearly presented, inducing curiosity and knowledge acquisition.
Easy to relate to personal consumption through a week rather than impact of single ingredient. Other factors than just co2.
it has the most information and it is very easy to use
Breaks down the impact of food on land
Gives the most complete picture. But for other uses cases I would pick Tool A.
Easiest use + easiest results
ease of use
They are very different, and made for different use-cases I assume. C is easy to use and understand (as opposed to the others which you would really work on for about 30minutes to really understand how your diet should potentially change, which is what I would use it for)
I think this was the tool that was most developed, at least looked like it.
Fun and a lot of information in one place
I can easily chose typical diets from the country I live in and then fine tune my answer which helps me get to my target faster.
It provides the most important information you need
Most interesting. Allows you to look at your diet from a different perspective.
I liked it more than tool A. Did not test tool B.

D.2 Particularly helpful features across tools

Was there a feature in any tool that you found particularly helpful?
If so, which one and why?

The midpoint indicators from Tool C was very nice
yes in tool C - as mentiond in my answers for tool C
Tool C had a nice overview of diets
Loved the suggested serving size in grams from A. So if that was included in C, as well as maybe compare the impact of the whole diet to different transportation methods, C would be perfect!
results are also describe in easier ways to make the impacts more visible
Sliders in Tool C
the overall diet and easy to slide the bars, might be hard to know how much portion you have in a week, and the green bar saying you are in the optimal average is introducing a bias since we want to put it there
The choice of units in tool A if the tool is for recipes.
Using an average person as a baseline in tool C.
The typical serving size in tool A.
Regionalization was a good thing. But also the message you get in Tool B that gives an idea of what it means was also good
The planetary boundaries. That was great! And how you could adjust the portions and see what happened. I also liked that it gave concreet feedback to what I could do to improve my diet. I also liked that it talked about per week, not just some random recipe
Presets for each country. Could be expanded with presets based on diet (eat alot of meat, vs vegetarian for example).
sliding scales, helps select numbers more easily. comparison with other emissions sources provides context.
Sliders, live-updated columns in Tool C, background knowledge and information hidden in tooltips.
Also took what was considered healthy into the calculations.
the results explained by type of product, by category and with resources to read more. Also they were given by country, that is so important
The carbon foot print of each individual item, thatz amazing.

In tool A when it suggested amounts for a normal serving.
How to enter ingredients on Tool A
The helpful tips on how to estimate how many grams of food on tool A
C provided a way to show what you should optimally eat, for health etc. The others just gave you a arbitrary number.
I liked the dragging lines in the tool C.
Tool A were you can compare ingredients
I also like the fact that the input methods are sliders and the tool also tells me what the ideal range for each item is.
Tool B is super simple and clean, so getting Tool C less crowded would help a lot.

D.3 Desired features for the future tool

What feature(s) were missing from any of the tools that you would like to see in a new environmental impact calculator?

AI to guess or fill in the recipe. F.e. write: Tik tok Pasta and it guesses the ingredients.
More meat related in calculator A
As written in the last one. Tool C with some of the features from A would be perfect!
alternative sources that can be adapted to minimise the impacts
Tracing impacts along supply chains
if you could for a week enter the ingredients you eat in a app and at the end it would show you the impact a bit like tool C it would be ideal
It does not provide the ability to save information about meals that I could find. It would be interesting to use it as a "diet" app where you track your calories for each meal over time, just for this case it would be for impacts instead. But it can also include information about health.
If human health is included, changing the input of kcals per day would be useful. Because right now the optimal diet seems to be between 2000-2500 kcal which is quite restrictive. Having the possibility to get comparative results side by side: like your own diet versus the one from another country (or another diet). Or comparing recipes.
The recipe based calculators would benefit from some pre-calculated recipes that can be adapted. E.g. start with the usual ingredients for pizza, then change the topping to see the difference.
Diverse impact categories
B was just a worse version of A in my opinion. The benefit was that it allowed you to add more details, but that also made it harder to implement the food you wanted to check. It also had less to compare it to. A was visually very pleasing, but by only giving feedback on the emissions made it hard to get any useful information out of it, unless you tried a bunch of combinations. C was great. It gave a lot of insight and it was very easy to use it to make relevant changes in your diet. But it required way more reading, and I wish they had used mass rather than portions, as that would have made it way better imo.
Open and disaggregated results to see the hotspots
Tool C, with better indication of amount, and also a real life example of CO2 emissions would be perfect tool.
Uncertainty ranges, more linkages to imported impacts.

maybe a section with premade meals with the option to edit the ingredients. In that way the whole process can be less tiring. it could be with just some basic dishes: pasta, pizza, sándwich, smoothies, and some typical dishes from each country.
Packaging costs
Include more environmental issues like bombs, unnecessary rocket for pleasure, huge production of electric cars etc.
Visualization of the impact beyond only numbers. Maybe something to visualize distance of a car or how many cars could be driven a kilometre.
Entering ingredients part of Tool A should be in Tool C and I would be happy.
Maybe make it more visual. Like drag and drop different foods onto a plate and it calculates for you
I like tool C, though it would be great if that was broken down into certain sample dinners/lunches, and show that what you tweak on the sliders also makes an impact on what vitamins/minerals you get, and what you end up emitting as a result. I like how playing around with these sliders can help argue that a given guy needs to eg. eat less meat and more fish. For that argument to hold up, the tool needs to show both an overview over what he would typically eat, and how that results in a certain emission and certain nutritional value.
maybe it would be nice to see a type of general values, like how is my impacts compared to the mean values
Explanations of why some things have more impact than others
Icons for pictures for the food item, say for example, pairing icon or a small image for fish next to the text "fish" would help in visual aid.
Option to save if wanted. To see improvement. And motivation to set goals for future diet.

E Interface and features results

What user interfaces or result presentations do you feel are missing and would like to see in the future?

Tool C graphs and percentages when clicking was great! Perfect with comparisons to real-world activities as in A as well!
nothing much
interactive ones
More sustainable ingredient alternatives should also evaluate health implications, so that the calculator does not suggest a more sustainable ingredient that is worse for your health.
As mentioned in a previous answer, side by side comparison of results. As suggested above, if there is an offer for more sustainable alternatives, it could be: if you replace X amount of meat by lentils, here is how your new diet would look like.
Some images could be useful: for example comparing one portion of red meat = 5 portions of chicken etc. Sometimes images convey messages in a better way. Cute emoticons would make the tool attractive.
Would be nice to be able to navigate through individual ingredients and see more informations about their impacts (e.g, https://impactco2.fr/outils/fruitsetlegumes).
It was not possible to choose between organic and non-organic food. Maybe something related to packaging could be useful as well: especially plastics.
A separate search bar or other indication of being able to just type in the ingredients instead of choosing them from a dropdown.
Maps of emissions locations.

Add childrens icon/image/cartoon so that it will show the use that all this effort is to have a stressfree environment for your future generation.
The list of food items could be more detailed, but you got to set the limit somewhere of course :)
The one previously mentioned, visual approach when adding food input.
if the recipe has moderate to high impact on the environment it would be a good to suggest frequency of intake for such recipe, such that it brings the impact within tolerable range. For example, say a pizza has an impact of xyz grams, but if you eat this only twice a month or for once every two months, you can bring the impact down by this much (some metric)
Would be cool if all online recipes in the future already had the impact calculated next to it. Or that it was easy to download it or copy/paste into a calculator (less than 1 minute work)

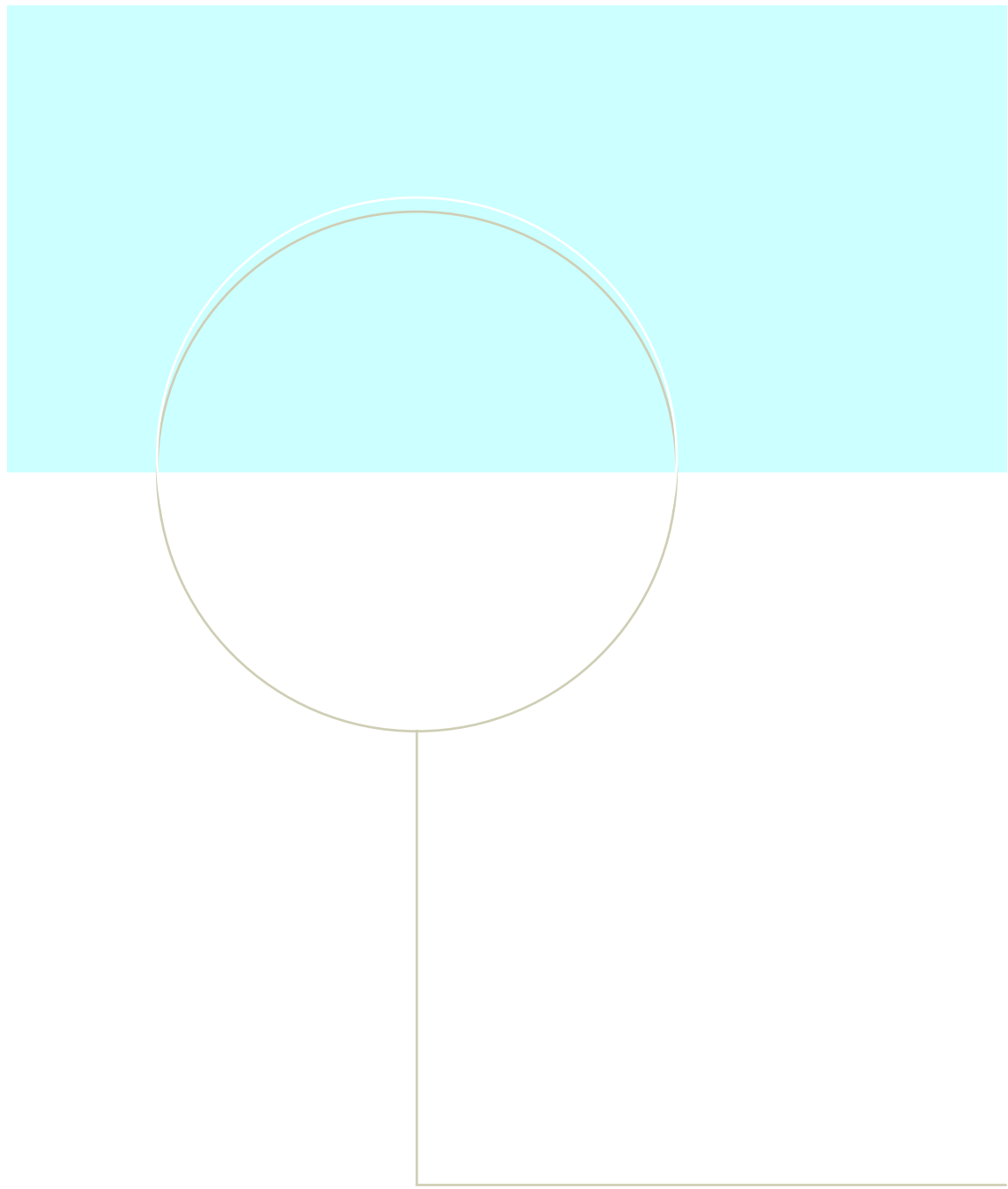
F Additional comments and suggestions

Do you have any other comments or suggestions about the calculators or the survey?

no
It would be interesting to write shopping list and see the impact of all the products I am planning to buy that week, and evaluate it together with health benefits. Maybe if the products are tracked over time and you can press on an option to "switch to more sustainable option" it can gather this information to say something about "this week you lowered your emissions by xx % by choosing more sustainable ingredients"
I rarely use calculators as just calculating GHG emissions is not something I am particularly interested in and sometimes I feel that I do not learn much from it. But an interactive tool with maybe a possibility to get to read more about individual food items or impacts could be useful.
Great project!
Focus on making the numbers matter to people who just want to see how they can improve or how they are doing in terms of sustainability. That's why C was my favorite.

If it comes as a mobile app, or integrated in the grocery shopping apps, it will be great.

Sad I could not open tool B, maybe its my safaris fault though...
Would be great to see planetary boundaries calculator for investment portfolios!
The link to tool B, don't know if its just me. Good luck!



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